

Interest Rate Risk around Monetary Policy Announcements and Asset Duration*

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Abstract

This paper examines the role of interest rate risk and asset duration in the monetary policy announcement premium. First, the average returns of long-duration Treasury bonds are higher than those of short-duration bonds on monetary policy announcement days. Second, whereas short-duration equities earn higher average monthly returns than long-duration equities, their announcement-day returns are similar to or lower than those of long-duration equities, depending on the duration measure. Third, the sensitivity of returns to changes in interest rate risk is greater for long-duration assets than for short-duration assets, a result that holds for both bonds and equities. These findings support an interest-rate-risk channel in which declines in interest rate risk on monetary policy announcement days raise the returns of long-duration assets.

Keywords: Monetary policy announcements; interest rate risk; asset duration; term structure of returns

JEL Classification: E43, E52, G12, G14

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1 Introduction

This paper examines the monetary policy announcement premium by exploiting the heterogeneity in asset duration and exposure to interest rate risk. The monetary policy announcement premium refers to the empirical pattern in which the average asset returns are elevated on announcement days; the return on the S&P 500 index is 25.5 basis points on Federal Open Market Committee (FOMC) days and 2.7 basis points on non-FOMC days ([Savor and Wilson, 2013](#)). Prior to announcements, investors face greater risk than in normal periods and demand higher expected returns to hold assets. An important question is what type of risk investors face prior to monetary policy announcements. The main findings of this paper are as follows: (i) a decline in market-implied interest rate risk is associated with higher asset returns and (ii) asset duration is a key determinant of cross-sectional heterogeneity in risk exposures.

First, this paper examines the role of interest rate risk in the announcement premium. Empirical evidence suggests that monetary policy announcements convey information about the future paths of interest rates, thereby reducing interest rate risk, as reflected in the decline in option-implied interest rate volatility around announcements ([Bauer, Lakdawala, and Mueller, 2022](#)). While the existing literature proposes uncertainty-based mechanisms for explaining the announcement premium ([Ai and Bansal, 2018](#); [Wachter and Zhu, 2022](#)), this paper further investigates the particular type of uncertainty surrounding monetary policy announcements.

Second, assets with different durations exhibit different exposures to interest rate risk. When interest rates rise unexpectedly, long-duration assets decline in value more than short-duration assets do, as future cash flows are discounted at higher discount rates. Conversely, a decline in interest rates leads to greater gains for long-duration assets. Prior to announcements, investors who prefer early resolution of uncertainty assign lower valuations to long-duration assets. Short-duration assets are less sensitive to interest rate changes because of their near-term cash flows.

We develop a tractable asset pricing model in which a representative investor with recursive preferences allocates wealth between short- and long-duration assets under uncertainty. The model focuses on discount factor, cash-flow, and consumption risks, which are resolved by monetary policy announcements. The distinction between short- and long-duration assets lies in the timing of their consumption claims. Because long-duration assets represent claims on consumption in the future, they are more exposed to discount factor risk.

The model clarifies why asset duration is useful for identifying an interest-rate-risk channel in announcement returns. Interest-rate risk affects both the valuation of future consumption and the discounted value of future asset payoffs. Discount-factor uncertainty contributes to announcement returns through a variance term whose importance increases with asset duration. The model shows analytically that long-duration assets are therefore more exposed to discount-factor risk.

For bonds with predetermined cash flows, this duration effect implies that, when announcement-related consumption risk is sufficiently small, long-duration bonds earn higher announcement returns and respond more strongly to the amount of interest-rate risk resolved by the announcement. For equities, stochastic cash flows introduce additional covariance terms that can amplify or offset this channel, making both the return ordering and the duration gradient in sensitivity theoretically ambiguous. This distinction motivates our empirical analysis: we test the model’s prediction for bonds and examine whether the same stronger exposure of long-duration assets to market-implied interest rate risk emerges empirically for equities.

We empirically analyze the expected returns and return sensitivities to interest rate risk across asset durations. The analysis focuses on U.S. Treasury bonds and publicly traded U.S. firms. We measure the equity duration using three approaches commonly employed in the literature: [Weber \(2018\)](#), [Hansen, Heaton, and Li \(2008\)](#), and [Gormsen and Lazarus \(2023\)](#).

We document that longer-duration bonds have higher average returns than short-duration bonds on FOMC days. For example, five-year Treasury bonds earn an average return of 4.4 basis points on monetary policy announcement days, while twenty-year bonds earn an average return of 10.1 basis points. This supports the view that long-duration bond returns are more sensitive to discount factor risk.

For equities, FOMC announcements tilt returns toward long-duration firms relative to their monthly return patterns. Using cash-flow-based durations, average announcement-day returns are nearly flat: firms with durations of 14 and 22 years earn 24.4 and 25.4 basis points, respectively, while other duration measures produce more upward-sloping patterns. A particularly robust result comes from comparing FOMC-day returns with monthly returns: unlike on announcement days, short-duration equities earn higher average monthly returns. This contrast is consistent with the interest-rate-risk-resolution channel. Because FOMC announcements resolve more interest rate uncertainty than ordinary days and long-duration equities are more sensitive to discount-rate news, their relative performance improves around announcements even when the unconditional term structure of equity returns slopes downward.

We also show that the sensitivity of returns to changes in interest rate risk increases with duration for both bonds and equities. We measure interest rate risk using the option-implied volatility of Eurodollar futures ([Bauer, Lakdawala, and Mueller, 2022](#)). For bonds, a 1% decline in interest rate risk on FOMC announcement days raises the return on 10-year bonds by 10.4 basis points, compared with 6.1 basis points for 5-year bonds.

This duration gradient is also evident among equities. For equities with a duration of 23 years, the sensitivity of returns to interest rate risk is 18, implying an 18-basis-point increase in returns following a 1% decline in interest rate risk on announcement days. For equities with a duration of 9 years, the corresponding sensitivity is only 9. Thus, for both bonds and equities, longer-duration

assets respond more strongly to the resolutions of interest rate risk.

The remainder of this paper is organized as follows. Section 2 presents a simple theory. Section 3 describes the data and variable construction. Section 4 empirically analyzes bonds and Section 5 conducts the corresponding analysis for equities. Section 6 concludes the paper.

Related Literature. First, this paper focuses on interest rate risk as an additional mechanism underlying the premium, while much of the existing literature emphasizes cash-flow risk. Empirically, [Lucca and Moench \(2015\)](#) and [Hu et al. \(2022\)](#) show that excess returns are higher when aggregate uncertainty, commonly proxied by the VIX, falls substantially on FOMC announcement days. [Zhang and Zhao \(2023\)](#) examine the contemporaneous relationship between reductions in the VIX and the announcement premium.¹ Theoretically, models such as [Wachter and Zhu \(2022\)](#) and [Ai et al. \(2022\)](#) primarily attribute the announcement premium to cash-flow risk. By introducing a framework that incorporates both interest rate risk and cash-flow risk, this paper provides a complementary perspective on the risks resolved through monetary policy announcements.

In addition to interest rate risk, this paper highlights duration as a key source of cross-sectional heterogeneity in exposure to monetary policy announcements. Prior empirical studies have often tested risk-based hypotheses by examining cross-sectional variations in asset sensitivities, such as the relationship between the market beta and expected returns ([Savor and Wilson, 2014](#); [Wachter and Zhu, 2022](#)), or the predictive power of option-implied variance reductions for excess returns ([Ai et al., 2022](#)). Theoretically, prior studies have generally treated cross-sectional heterogeneity in sensitivity as exogenous and have not directly addressed its sources ([Wachter and Zhu, 2022](#)). This paper interprets the underlying determinants of such heterogeneity, demonstrating that variation in exposure to monetary policy announcements is systematically driven by differences in asset durations.

Second, this paper contributes to the literature that emphasizes the role of monetary policy uncertainty in determining asset prices. Prior research has focused on average market responses such as movements in yields and stock indices, with limited attention paid to cross-sectional heterogeneity.² This paper addresses this gap by focusing on the cross-sectional heterogeneity in asset duration.

¹Related empirical contributions include [Brusa, Savor, and Wilson \(2020\)](#); [Neuhierl and Weber \(2018\)](#); [Cieslak, Morse, and Vissing-Jorgensen \(2019\)](#); [Mueller, Tahbaz-Salehi, and Vedolin \(2017\)](#); [Indriawan, Jiao, and Tse \(2021\)](#); [Wachter and Zhu \(2022\)](#).

²[Bundick, Herriford, and Smith \(2017\)](#) estimate that changes in short-term uncertainty positively impact the term premium on announcement days. [Bauer, Lakdawala, and Mueller \(2022\)](#) highlight that reductions in uncertainty influence asset prices in ways distinct from conventional policy surprises. [Lakdawala, Moreland, and Schaffer \(2021\)](#) demonstrate that changes in uncertainty affect spillovers to global bond yields. Similarly, [Kroencke, Schmeling, and Schrimpf \(2021\)](#) identify shifts in risk appetite on FOMC announcement days and show that these shifts are correlated with stock returns.

Third, this paper contributes to the literature on the term structure of asset returns. [Lettau and Wachter \(2011\)](#), [Van Binsbergen and Koijen \(2017\)](#), and [Weber \(2018\)](#) document a downward-sloping term structure of monthly average equity returns. This paper extends this evidence by examining the term structure specifically on monetary policy announcement days rather than using monthly averages. The results show that the term structure on announcement days differs from the pattern observed for monthly average returns.

2 Model

This section develops a simple model to provide intuition on the announcement premium and its cross-sectional variation across asset durations. The framework builds on [Ai and Bansal \(2018\)](#) and introduces two key extensions: (i) investors face discount factor risk, and (ii) the model incorporates asset-duration heterogeneity.

There is a representative investor in a four-period model. Periods 1 and 2 are designated as trading periods, whereas periods 2, 3, and 4 are consumption periods. The investor can trade two types of assets: short-duration assets, which provide claims for consumption in period 3, and long-duration assets, which provide claims for consumption in period 4.

In period 1, the investor faces three sources of uncertainty: discount factor, cash-flow risk, and consumption risks. Let $\mathcal{S} = \{s_1, \dots, s_N\}$ denote the finite set of states, one of which is revealed through the announcement. The asset market opens before the state is realized, so period-1 choices are not indexed by s . At the beginning of period 2, an announcement reveals the true state $s \in \mathcal{S}$, which is then known to agents in periods 2, 3, and 4. After this uncertainty is resolved, a second asset market opens, and period-2 prices and choices are state contingent. In periods 3 and 4, the investor consumes the returns from these assets.

The investor derives utility from consumption in periods 2, 3, and 4 and maximizes the expected utility:

$$\max_{\theta_1^S, \theta_1^L, B_1, \{\theta_2^S(s), \theta_2^L(s)\}_{s \in \mathcal{S}}} \left\{ E_1 \left[\left(C_2^{1-\frac{1}{\psi}} + \beta(s)C_3(s)^{1-\frac{1}{\psi}} + \beta(s)^2C_4(s)^{1-\frac{1}{\psi}} \right)^{\frac{1-\gamma}{1-\frac{1}{\psi}}} \right] \right\}^{\frac{1}{1-\gamma}},$$

subject to

$$e = P_1^S \theta_1^S + P_1^L \theta_1^L + B_1,$$

$$B_1 = C_2 + P_2^S(s) \theta_2^S(s) + P_2^L(s) \theta_2^L(s),$$

$$C_3(s) = X_3(s)(\theta_1^S + \theta_2^S(s)),$$

$$C_4(s) = X_4(s)(\theta_1^L + \theta_2^L(s)), \quad s \in \mathcal{S},$$

where e is the initial endowment and B_1 denotes wealth carried from period 1 to period 2. The variables θ_1^S and θ_1^L are the period-1 holdings of the short- and long-duration assets, chosen before the state is revealed. After the announcement at the beginning of period 2, the investor chooses state-contingent holdings $\theta_2^S(s)$ and $\theta_2^L(s)$. The payoffs $X_3(s)$ and $X_4(s)$ are the cash flows of the short- and long-duration assets in periods 3 and 4, respectively.

The announcement valuation term is defined as

$$H(s) \equiv \left[C_2^{1-\frac{1}{\psi}} + \beta(s)C_3(s)^{1-\frac{1}{\psi}} + \beta(s)^2C_4(s)^{1-\frac{1}{\psi}} \right]^{\frac{\frac{1}{\psi}-\gamma}{1-\frac{1}{\psi}}},$$

which is analogous to the announcement stochastic discount factor in [Ai and Bansal \(2018\)](#). The expected announcement returns on the short- and long-duration assets can then be written as

$$R^S \equiv E_1 \left[\frac{P_2^S(s)}{P_1^S} \right] = \frac{E[H(s)] E[\beta(s)X_3(s)C_3(s)^{-1/\psi}]}{E[H(s)\beta(s)X_3(s)C_3(s)^{-1/\psi}]},$$

$$R^L \equiv E_1 \left[\frac{P_2^L(s)}{P_1^L} \right] = \frac{E[H(s)] E[\beta(s)^2X_4(s)C_4(s)^{-1/\psi}]}{E[H(s)\beta(s)^2X_4(s)C_4(s)^{-1/\psi}]}.$$

Announcement returns depend on the covariance between $H(s)$ and the corresponding discounted future payoff. In the model, C_2 is non-stochastic because period 2 represents the immediate post-announcement period. This assumption is made for expositional clarity, and relaxing it does not alter the main results.

2.1 Bonds as a Case without Cash-Flow Uncertainty

We consider bonds with deterministic cash flows, so that $X_t(s)$ does not vary across states. In this case, the expected returns on the short- and long-duration bonds are given by

$$R^S \simeq 1 + \left(\gamma - \frac{1}{\psi} \right) \text{Cov} \left(\widehat{H}, \widehat{\beta} - \frac{1}{\psi} \widehat{c}_3 \right),$$

$$R^L \simeq 1 + \left(\gamma - \frac{1}{\psi} \right) \text{Cov} \left(\widehat{H}, 2\widehat{\beta} - \frac{1}{\psi} \widehat{c}_4 \right). \quad (1)$$

Here, hats on primitive variables denote the log deviations from unconditional means, while $\hat{H}$ denotes the first-order state-dependent component of the announcement valuation term defined above.

$$\hat{H}(s) \equiv \frac{a_3 + 2a_4}{1 - \frac{1}{\psi}} \hat{\beta}(s) + a_3 \hat{c}_3(s) + a_4 \hat{c}_4(s).$$

The parameters a_3 and a_4 are utility-weighted consumption shares for periods 3 and 4, respectively. Their precise definitions and the derivation of (1) are provided in Appendix A.

A positive announcement premium in the model corresponds to

$$\left(\gamma - \frac{1}{\psi}\right) \text{Cov} \left(\hat{H}, \hat{\beta} - \frac{1}{\psi} \hat{c}_3 \right) > 0, \quad \left(\gamma - \frac{1}{\psi}\right) \text{Cov} \left(\hat{H}, 2\hat{\beta} - \frac{1}{\psi} \hat{c}_4 \right) > 0. \quad (2)$$

The economic intuition behind (2) follows from the covariance between the asset valuation term and future payoff. The term $\hat{H}$ summarizes the first-order state variation in the announcement valuation term and reflects the discount factor and future consumption. The term $\hat{\beta} - \frac{1}{\psi} \hat{c}$ reflects the future payoff that is determined by the discount factor and the marginal utility of future consumption.

Economically, the discount-factor component captures the resolution of interest-rate uncertainty at the announcement. Before the announcement, uncertainty about future short-term rates generates uncertainty about asset valuations and, when investors value early resolution of uncertainty, lowers pre-announcement prices. The announcement reveals information about future rates, generating positive announcement returns.

Future consumption provides an additional general-equilibrium channel. Monetary policy can affect consumption through intertemporal substitution, while the announcement can also reveal information about future economic fundamentals (Nakamura and Steinsson, 2018; Cieslak and Pang, 2021). Consumption news therefore affects both $\hat{H}$ and the marginal utility terms in post-announcement prices and can either amplify or offset the direct discount-factor channel. As shown below, when the consumption risk resolved by the announcement is sufficiently small relative to discount-factor risk, the direct duration channel dominates.

Return Difference. Equation (1) makes the duration channel transparent. The post-announcement value of the long-duration bond loads twice as heavily on discount-factor news as that of the short-duration bond: the relevant covariance term contains $2\hat{\beta}$ for the long-duration bond, compared with $\hat{\beta}$ for the short-duration bond. Thus, the long-duration bond is more exposed to the discount-factor risk resolved at the announcement.

Consumption fluctuations can modify this comparison because future consumption also enters the covariance terms through marginal utility. When consumption is sufficiently smooth, however, these consumption-related terms are small, and the difference in exposure to $\hat{\beta}$ dominates. In this

case, the resolution of discount-factor uncertainty generates a larger announcement premium for the long-duration bond.

Proposition 1. *Suppose that $\left(\gamma - \frac{1}{\psi}\right) \frac{1}{1-\frac{1}{\psi}} > 0$. Then the contribution of discount-factor risk, $\text{Var}(\hat{\beta})$, to the announcement premium of the long-duration bond is twice as large as that of the short-duration bond. Moreover, if the consumption risk resolved by the announcement is small relative to discount-factor risk, then*

$$R^L > R^S > 1.$$

Proof. See Appendix A.2. □

Declines in Interest Rate Risk. We next examine how announcement returns respond to the amount of interest rate risk resolved by the announcement. Write $\hat{\beta} = \sigma_\beta z_\beta$, $\text{Var}(z_\beta) = 1$, so that σ_β indexes the amount of discount-factor risk resolved at the announcement. Because the long-duration bond loads more heavily on $\beta(s)$, an increase in σ_β has a stronger direct effect on its announcement return. Stochastic consumption can modify this comparison through marginal-utility effects. When the consumption risk resolved by the announcement is sufficiently small relative to discount-factor risk, however, the direct duration channel dominates.

Proposition 2. *Suppose that bond cash flows are deterministic and that $\left(\gamma - \frac{1}{\psi}\right) \frac{1}{1-\frac{1}{\psi}} > 0$. If the consumption risk resolved by the announcement is sufficiently small relative to discount-factor risk, then*

$$\frac{\partial R^L}{\partial \sigma_\beta} > \frac{\partial R^S}{\partial \sigma_\beta}.$$

Thus, resolving more interest rate risk raises the announcement return of the long-duration bond more than that of the short-duration bond.

Proof. See Appendix A.3. □

The smooth-consumption condition provides a sufficient condition for a stronger response of long-duration bond returns, but it is not necessary for the empirical analysis. The relative response of long- and short-duration bonds depends on the joint behavior of discount-factor and consumption news and is therefore an empirical matter.

2.2 Equities with Cash-Flow Uncertainty

We consider equities with stochastic cash flows. The announcement premium of short- and long-duration equities is given by

$$\begin{aligned}
R^S &\simeq 1 + \left(\gamma - \frac{1}{\psi}\right) \text{Cov} \left(\widehat{H}, \widehat{\beta} + \widehat{x}_3 - \frac{1}{\psi} \widehat{c}_3 \right), \\
R^L &\simeq 1 + \left(\gamma - \frac{1}{\psi}\right) \text{Cov} \left(\widehat{H}, 2\widehat{\beta} + \widehat{x}_4 - \frac{1}{\psi} \widehat{c}_4 \right). \\
R^L - R^S &\simeq \left(\gamma - \frac{1}{\psi}\right) \text{Cov} \left(\widehat{H}, \widehat{\beta} + \frac{1}{\psi} (\widehat{c}_3 - \widehat{c}_4) - \widehat{x}_3 + \widehat{x}_4 \right). \tag{3}
\end{aligned}$$

The derivation is provided in Appendix A. The additional covariance term that is absent in the bond case is $\text{Cov}(\widehat{H}, \widehat{x})$, which arises from stochastic cash flows. Unlike the bond case, the duration premium for equities is not determined only by discount-factor risk. It also depends on whether cash-flow news is more strongly related to discount-factor news and consumption news at short or long horizons.

Proposition 3. *Suppose that cash flows are stochastic and consumption risk is small relative to discount-factor risk. Then, to first order in consumption risk, the long-short announcement return differential is determined by*

$$\text{Var}(\widehat{\beta}) + \text{Cov}(\widehat{\beta}, \widehat{x}_4) - \text{Cov}(\widehat{\beta}, \widehat{x}_3).$$

Thus, unlike bonds, smooth consumption alone does not imply that long-duration equities earn higher announcement returns.

Proof. See Appendix A.4. □

These covariance terms between discount factor and cash-flow capture the extent to which monetary policy announcements affect expected future cash flows as well as discount rates. In this sense, the model allows monetary policy news to operate through the cash-flow channel emphasized in the existing literature (Cieslak and Pang, 2021; Cieslak and McMahon, 2023). The relative importance of this channel for long- and short-duration equities depends on how cash-flow news co-moves with discount-factor news across horizons.

Declines in Interest Rate Risk. Finally, we derive the equity counterpart to Proposition 2. Under the small-consumption-risk condition, the effect of the amount of discount-factor risk resolved by the announcement is, to leading order,

$$\frac{\partial(R^L - R^S)}{\partial\sigma_\beta} \simeq \left(\gamma - \frac{1}{\psi}\right) \frac{a_3 + 2a_4}{1 - \frac{1}{\psi}} [2\sigma_\beta - \rho_{\beta x_3}\sigma_{x_3} + \rho_{\beta x_4}\sigma_{x_4}]. \tag{4}$$

Its derivation is in Appendix A.5.

Unlike the bond case, equity returns contain the additional cash-flow term $-\rho_{\beta x_3}\sigma_{x_3} + \rho_{\beta x_4}\sigma_{x_4}$. This term allows the model to capture the possibility that monetary policy announcements affect equity valuations not only through discount rates but also through news about expected future cash flows (Lettau and Wachter, 2007; Cieslak and Pang, 2021; Cieslak and McMahon, 2023). Accordingly, the relative response of long- and short-duration equities depends on how cash-flow news at different horizons covaries with discount-factor news. Whether long-duration equities respond more strongly to the resolution of interest rate risk is therefore an empirical matter.

3 Data

This section details the construction of the main variables: bond and equity returns, equity duration, and interest rate risk. The sample period runs from the first quarter of 1994 to the fourth quarter of 2019, when data on interest rate risk are available.

3.1 Interest Rate Risk

The measure of interest rate risk is constructed using the market-based conditional volatility of the future short-term interest rate, as proposed by Bauer, Lakdawala, and Mueller (2022). Specifically, Bauer, Lakdawala, and Mueller (2022) compute the standard deviation of one-year-ahead Eurodollar (ED) futures, conditional on current information, denoted by $IRU_t \equiv \sqrt{\text{Var}(\text{ED}_{t+\tau} | I_t)}$. This approach yields a model-free estimate of the conditional standard deviation using information embedded in futures and options prices.

Figure 1 illustrates our measure of interest rate uncertainty around FOMC announcements. Figure 1(a) plots the distribution of the daily log changes in the uncertainty measure for FOMC and non-FOMC days. The distribution for FOMC days shifts to the left relative to that of non-FOMC days, indicating that interest rate uncertainty tends to decline more on FOMC announcement days than on non-FOMC days. Figure 1(b) shows the timing of the decline by plotting the average daily changes around FOMC days. The decline is sharply concentrated on the FOMC announcement day (shown as zero in the figure), whereas average changes are close to zero before and after the announcement. These patterns support the interpretation that the measured decline in uncertainty is driven by information released during FOMC announcements, rather than by ordinary daily fluctuations.

Figure 1(a) also shows a substantial variation in the change in interest rate risk, even though the average change is negative. This variation is often related to changes in the Federal Reserve's

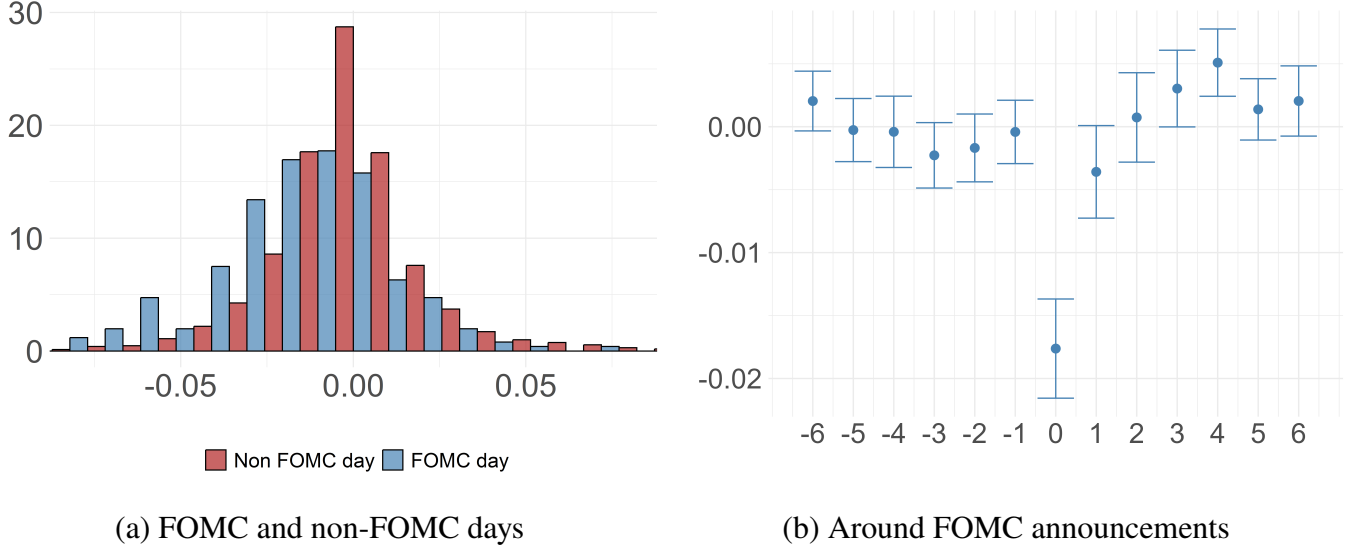


Figure 1: Changes in Interest Rate Risk around FOMC Announcements

Note: Panel (a) plots the distribution of daily log changes in the interest rate uncertainty measure separately for FOMC and non-FOMC days. Panel (b) plots the average daily log change in the uncertainty measure from six business days before to six business days after FOMC announcements. Day 0 denotes the FOMC announcement day. Two-standard-error bands are also shown.

forward guidance (Lakdawala, Moreland, and Schaffer, 2021).³ In the subsequent analysis, we exploit the variation in interest rate risk to estimate the sensitivity of asset returns with respect to these changes. Conceptually, the empirical approach compares asset returns across announcement days that differ in the magnitude of the change in interest rate risk.

A potential concern with this measure is that the relevant option market may not be sufficiently deep, so option-implied uncertainty could contain noise unrelated to changes in uncertainty induced by FOMC announcements. To address this concern, we construct a principal component measure using several interest-rate uncertainty proxies.⁴ This method follows Nakamura and Steinsson (2018), who construct the first principal component of multiple futures contracts. This principal component measure mitigates the concern that our results are driven by idiosyncratic noise in any

³For example, the largest decrease occurred in August 2011; interest rate uncertainty decreased by 29%. Before the meeting, the FOMC stated that interest rates would be kept low “... for an extended period.” At the August meeting, the FOMC explicitly signaled that rates would remain low “at least through mid-2013.” The market can interpret statements with less uncertainty about future interest rates. The central bank’s guidance plays a crucial role in reducing interest rate risk.

⁴Specifically, we use the change in our baseline interest-rate uncertainty measure, the change in basis-point volatility, defined as the product of Black implied volatility and the futures price and computed from at-the-money Eurodollar options, and the change in the MOVE index, which is a weighted average of basis-point volatility for one-month Treasury options across maturities. We restrict the sample to FOMC announcement days and compute changes in each measure around the announcement window. We then extract the first principal component after centering and standardizing the three variables.

particular option-based measure or market.

Relation to Monetary Policy Surprises. We briefly discuss the relationship between interest rate uncertainty and conventional monetary policy shocks. Figure 2 plots the change in interest rate uncertainty against the monetary policy shock in Nakamura and Steinsson (2018). The two measures are positively related: a simple regression of the announcement change in interest rate uncertainty on the monetary policy shock also yields a statistically significant slope.

However, the two measures only partially overlap. The regression has an R^2 value of 0.12, implying that a monetary policy shock explains 12 percent of the variation in interest rate uncertainty across announcements. Equivalently, the remaining 88 percent of the variation is not explained by this monetary policy surprise measure in the linear regression. Therefore, while changes in interest rate uncertainty tend to comove with monetary policy shocks, the uncertainty measure contains variations beyond the standard monetary policy shock. We therefore control for the monetary policy shock, but the estimated effects of interest rate uncertainty do not capture exposure to the monetary policy shock. The estimation results confirm that adding the monetary policy shock to the empirical specification has only a small effect on the estimated coefficients.

3.2 The Duration of Assets

We discuss the construction of the asset durations. For zero-coupon bonds, duration equals the number of years to maturity; thus, a bond with n years to maturity has duration n .

Following Weber (2018), we construct equity duration based on the timing of expected cash flows, analogous to Macaulay duration for bonds. This approach reflects the weighted average time to maturity of cash flows, where the weights are determined by the ratio of discounted future cash flows to the current price:

$$\text{Duration}_{i,t} = \frac{\sum_{s=1}^{\infty} s \times \text{CF}_{i,t+s} / (1+r)^s}{P_{i,t}},$$

where $\text{Duration}_{i,t}$ is the duration of firm i at time t , $\text{CF}_{i,t+s}$ is cash flow at time $t+s$, $P_{i,t}$ is the current equity price, and r is the constant discount rate. Unlike bonds, equities lack observable finite maturities and predetermined cash flows. To account for this feature, the duration formula is decomposed into two components: a finite-horizon term of length T , and a residual term that captures the infinite-horizon component:

$$\text{Duration}_{i,t} = \frac{\sum_{s=1}^T s \times \text{CF}_{i,t+s} / (1+r)^s}{P_{i,t}} + \left(T + \frac{1+r}{r} \right) \frac{P_{i,t} - \sum_{s=1}^T \text{CF}_{i,t+s} / (1+r)^s}{P_{i,t}}. \quad (5)$$

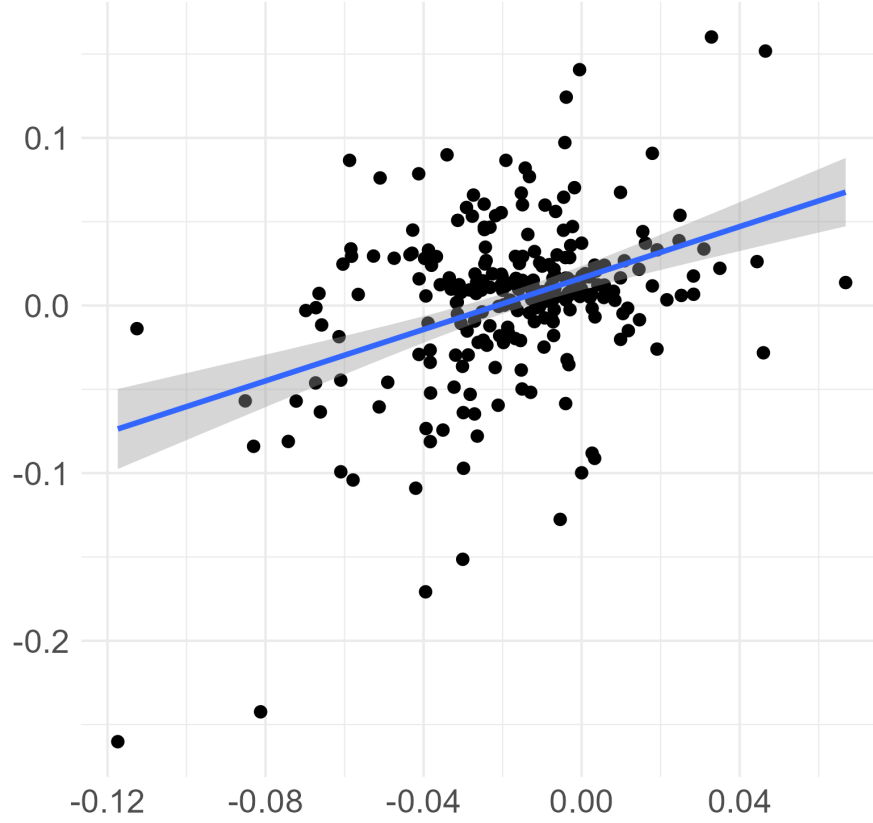


Figure 2: Interest Rate Uncertainty and Monetary Policy Shocks

Note: This figure shows the relationship between the announcement change in interest rate uncertainty and the monetary policy shock from [Nakamura and Steinsson \(2018\)](#). The horizontal axis is the change in interest rate uncertainty, and the vertical axis is the monetary policy shock. Each dot represents an FOMC announcement. The solid line is the fitted value from a linear regression, and the shaded area shows the 95% confidence interval.

Appendix [C.1](#) presents the derivation. Future cash flows are forecast using the projected return on equity and growth in book equity based on clean surplus accounting.

$$\begin{aligned}
 CF_{i,t+s} &= E_{i,t+s} - (BV_{i,t+s} - BV_{i,t+s-1}) \\
 &= BV_{i,t+s-1} \left[\frac{E_{i,t+s}}{BV_{i,t+s-1}} - \frac{BV_{i,t+s} - BV_{i,t+s-1}}{BV_{i,t+s-1}} \right].
 \end{aligned} \tag{6}$$

Return on equity, defined as $E_{i,t+s}/BV_{i,t+s-1}$, is the ratio of income before extraordinary items to lagged book equity. Both return on equity and growth in book equity are modeled as autoregressive processes, with parameters estimated using pooled data from CRSP-Compustat firms. Additional

Table 1: Summary Statistics and Correlations for Firm Characteristics.

	Dur	BM	Size	Prof	Lev	Sales	ME
Panel A. Means and Standard Deviations							
Mean	18.0	0.77	6.0	0.01	0.21	1.63	2539
SD	6.6	1.0	2.1	0.06	0.23	20.5	10179
Panel B. Correlations							
Dur		-0.61	-0.02	-0.10	-0.04	0.01	0.09
BM			-0.00	-0.07	-0.09	-0.05	-0.10
Size				0.23	0.26	-0.00	0.42
Prof					0.08	0.16	0.12
Lev						-0.00	0.06
Sales							0.00

Note: Table 1 reports time-series averages of quarterly cross-sectional means and standard deviations for firm characteristics in Panel A and correlations among these variables in Panel B. “Dur” is cash-flow duration in years; “BM” is the book-to-market ratio; “Size” is the log of assets; “Prof” is profit divided by assets; and “Lev” is the leverage ratio. “Sales” is percentage sales growth. “ME” is market capitalization in millions. Financial statement data come from quarterly Compustat. The sample period is 1994Q1–2019Q4.

details on the estimation procedure are provided in Appendix C.

Table 1 presents summary statistics for the key firm characteristics in Panel A. The average duration implied by stock prices is 18.0 years, with a standard deviation of 6.6 years, reflecting substantial heterogeneity across firms. On average, the public (19.8 years) and transportation (18.0 years) industries have longer durations, while the utilities (16.3 years) and wholesale (16.8 years) industries have shorter durations.

Panel B reports the cross-sectional correlations among the key firm characteristics. The book-to-market ratio is defined as the ratio of book equity to market equity, where book equity is calculated as total stockholders’ equity plus deferred taxes and investment tax credits, minus the book value of the preferred stock, and market equity is measured as total market capitalization.

Panel B shows a strong negative correlation between duration and the book-to-market ratio, with a correlation coefficient of -0.61. As shown in Appendix C.2, this theory implies a negative linear relationship between cash-flow duration and the book-to-market ratio. The close alignment between the theory and the data suggests that our measure of cash-flow duration supports the validity of our duration measure.

Following the literature that employs it as an alternative proxy for duration (Lettau and Wachter, 2011; Hansen, Heaton, and Li, 2008), we use the book-to-market ratio in robustness checks. We also use a duration measure based on Gormsen and Lazarus (2023).

3.3 Equity and Bond Returns

The equity sample consists of firms incorporated in the United States and listed on the NYSE, Amex, or Nasdaq, excluding those in the financial sector. Balance sheet information is obtained from the Compustat database. The daily return of an asset on announcement days is calculated as $\frac{p_t - p_{t-1}}{p_{t-1}}$, where p_t denotes the closing price on announcement day and p_{t-1} denotes the closing price on the previous day. For equities, daily closing price data are obtained from CRSP.

For Treasury bonds, returns are constructed using data from [Liu and Wu \(2021\)](#). This dataset provides the daily zero-coupon yields on U.S. Treasury bonds for various maturities $Y_t(n)$. The price of a bond with maturity n years, $p_t(n)$, is $p_t(n) = \frac{1}{(1+Y_t(n))^n}$, where $Y_t(n)$ is the annualized yield of a bond with maturity n years at time t .

4 Empirical Analysis of Bonds

4.1 Average Return on FOMC Days

This subsection empirically analyzes the average bond returns on FOMC days. Proposition 1 shows that long-duration bonds should exhibit higher returns than short-duration bonds. Figure 3(a) presents the average returns on FOMC days across maturities (in years), with 95% confidence intervals based on Newey–West standard errors. The results show a monotonic increase in returns with duration. For example, the average return on a one-year Treasury bond is 0.62 basis points, while that on a twenty-nine-year Treasury bond is 10.22 basis points. Figure 3(b) plots the intercepts from the regressions that control for the monetary policy shock and daily changes in the VIX. An upward term structure is still evident.

Long-duration bonds exhibit larger standard errors. This follows from the bond price formula, $p_t(n) = \frac{1}{(1+Y_t(n))^n}$, which implies that a given change in $Y_t(n)$ results in greater price variability for longer-maturity bonds. Consequently, the standard errors of the returns increase with duration. This pattern is consistent with that reported by [Wachter and Zhu \(2022\)](#).

We also regress the returns of Treasury bonds with different maturities on risk factors. The empirical specification is as follows:

$$y_t^m = \alpha^m + \beta_s^m X_{s,t} + \epsilon_t^m, \quad (7)$$

where y_t^m denotes the return on bonds with maturity m at time t , α^m is the pricing error, β_s^m is the time-series loading on risk factor s , and $X_{s,t}$ denotes the risk factor. Bond and market excess returns are expressed in basis points.

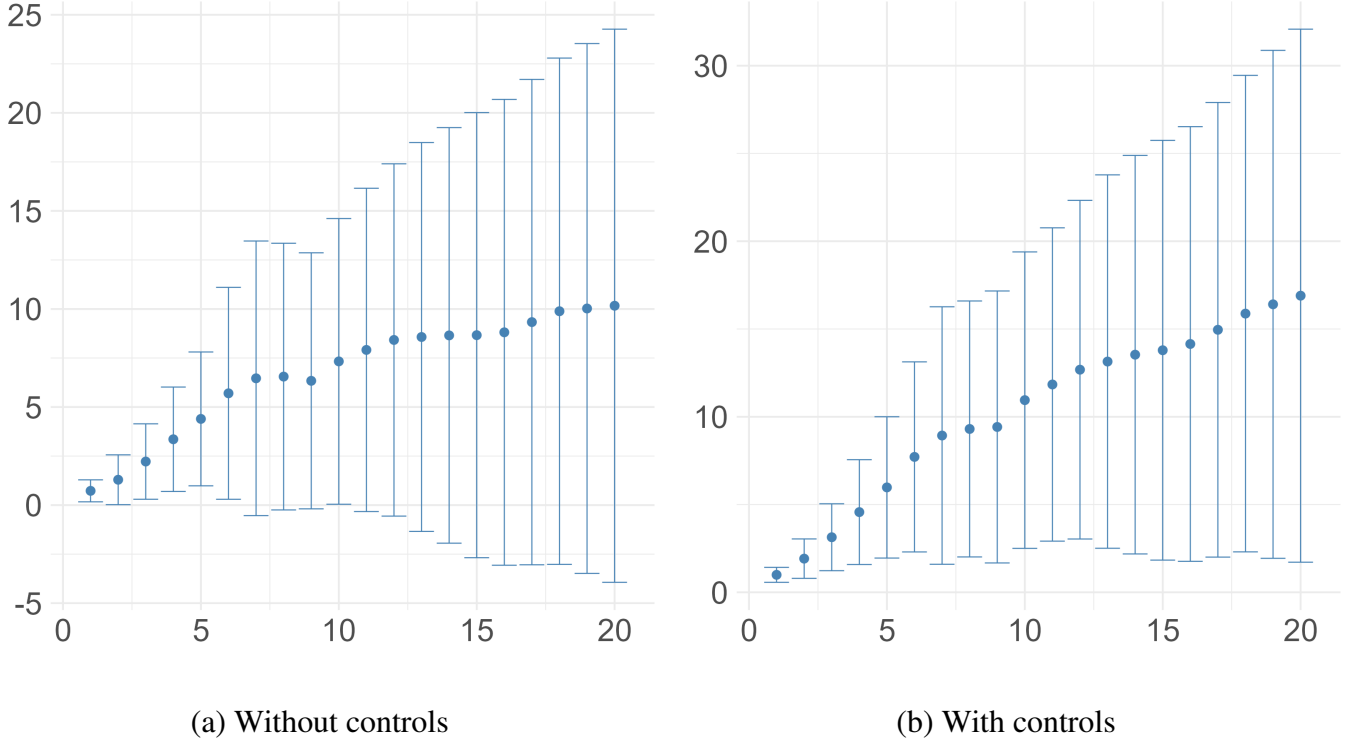


Figure 3: Average Returns on Treasury Bonds on FOMC Days.

Note: The vertical axis represents bond returns, expressed in basis points, and the horizontal axis represents bond duration in years. Panel (a) includes all FOMC days from 1994 to 2019. Panel (b) includes the monetary policy shock and the daily change in the VIX as controls. The 95% confidence intervals are calculated using Newey–West standard errors.

Table 2 reports the average returns on FOMC and non-FOMC days for Treasury bonds with maturities of 1, 5, 10, 15, and 20 years. Panel A presents the results for the FOMC days. On average, a long-short portfolio earns 9.4 basis points on FOMC announcement days. The table also reports the CAPM alphas and betas, where $X_{s,t}$ denotes the market excess return. Risk-adjusted returns, as measured by CAPM alphas, increase monotonically with maturity, from 0.7 for one-year bonds to 11.4 for twenty-year bonds.⁵

In contrast, returns on non-FOMC days are lower than those on FOMC days. For example, the average return for a one-year bond is 0.05 basis points on non-FOMC days, compared to 0.7 basis points on FOMC days. This is consistent with the view that interest rate risk is not substantially resolved on non-FOMC days. Moreover, the difference in average returns between short- and long-duration bonds is minimal on non-FOMC days, with a long-short portfolio earning only 1.7 basis points.

⁵ Lucca and Moench (2015) measure fixed-income yield changes from 2 p.m. on the preceding day to 2 p.m. on the FOMC day, before the announcement, and report no significant pre-announcement movements in fixed-income assets, whereas our close-to-close returns cover the full FOMC trading day.

Table 2: Average Returns on Treasury Bonds by Duration.

	Maturity				
	1	5	10	15	20
Panel A: FOMC days					
Average Return	0.7 (0.29)	4.4 (1.76)	7.3 (3.75)	8.6 (5.85)	10.1 (7.27)
α_{CAPM}	0.7 (0.37)	3.7 (2.13)	6.7 (3.79)	8.6 (5.68)	11.4 (7.34)
β_{CAPM}	0.00 (0.01)	0.02 (0.04)	0.02 (0.07)	0.01 (0.08)	-0.05 (0.10)
Panel B: Non-FOMC days					
Average Return	0.05 (0.05)	0.3 (0.33)	0.7 (0.62)	1.3 (0.90)	1.8 (1.14)
α_{CAPM}	0.07 (0.06)	0.4 (0.34)	1.0 (0.64)	1.7 (0.93)	2.3 (1.15)
β_{CAPM}	-0.01 (0.002)	-0.06 (0.008)	-0.11 (0.016)	-0.16 (0.023)	-0.20 (0.032)

Note: Table 2 reports the returns of Treasury bonds on FOMC days, time-series factor loadings (β), and pricing errors (α) for different maturities. The return on a Treasury bond is defined as $\frac{p_t(n) - p_{t-1}(n)}{p_{t-1}(n)}$, where $p_t(n)$ is the daily price of a Treasury bond with maturity n . Returns are stated in basis points. α_{CAPM} and β_{CAPM} are from the CAPM model. Newey–West standard errors are shown in parentheses. Maturity is in years.

4.2 Interest Rate Risk and Contemporaneous Regression

This section empirically investigates whether returns on bonds with longer durations are more responsive to changes in interest rate risk than those with shorter durations. This hypothesis relates to Proposition 2. For each bond duration m , we estimate the following time-series regression separately:

$$y_t^m = \alpha^m + \beta_{\text{IRU}}^m \Delta \text{IRU}_t + \epsilon_t^m, \quad (8)$$

where t indexes FOMC announcements and y_t^m denotes the return expressed in basis points, on a Treasury bond with duration m . ΔIRU_t is the percent change in interest rate risk from the previous day to the FOMC day. The coefficient β_{IRU}^m is estimated separately for each duration m .

Figure 4(a) shows the estimated values of β_{IRU}^m plotted against maturity, with 95% confidence intervals based on Newey–West standard errors. The coefficient decreases monotonically with duration, indicating that longer-duration bonds are more sensitive to changes in interest rate risk. For example, for a 10-year bond, β_{IRU}^m is -10.4, implying that a 1% decrease in implied interest rate volatility increases the return by 10.4 basis points, while for a 5-year bond, the coefficient is -6.1.

To distinguish the role of interest rate risk from that of broader measures of uncertainty, we

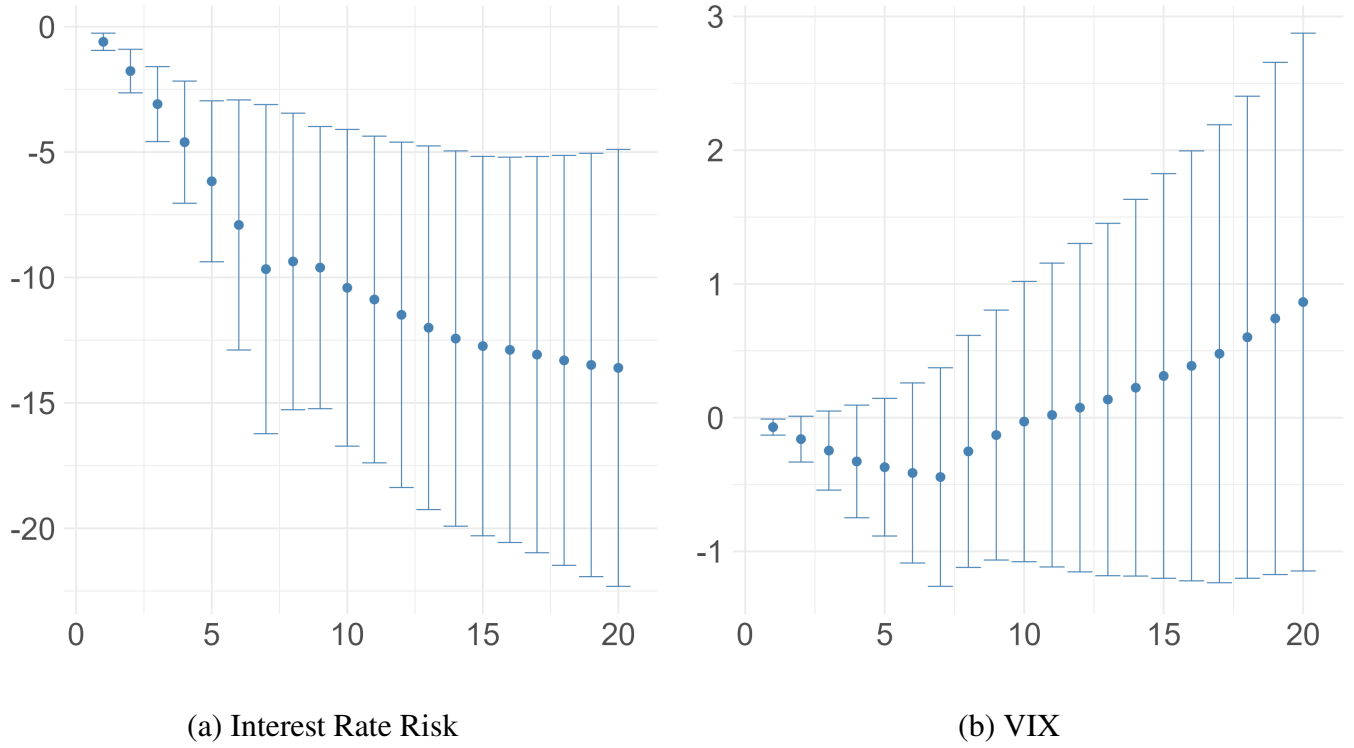


Figure 4: The Sensitivity of Returns to Changes in Uncertainty by Duration.

Note: The figure plots coefficients from regressions of Treasury returns on changes in the uncertainty measure:

$$y_t^m = \alpha^m + \beta^m \Delta \text{UNC}_t + \epsilon_t^m,$$

where ΔUNC_t represents changes in interest rate risk in Panel (a) and changes in the VIX in Panel (b). The vertical axis plots the coefficients β^m for each maturity m , along with their 95% confidence intervals.

examine the relationship between the VIX and bond returns. Specifically, we estimate the following:

$$y_t^m = \alpha_{vix}^m + \beta_{vix}^m \Delta \text{VIX}_t + \epsilon_t^m, \quad (9)$$

where ΔVIX_t is defined as the percent change in the VIX from the previous day to the FOMC day.

Figure 4(b) shows that the estimated values of β_{vix}^m are not statistically different from zero and do not exhibit a clear pattern with duration. For example, the coefficient is -0.4 for a 5-year bond and -0.1 for a 10-year bond. This suggests that Treasury bonds are not significantly exposed to VIX-related risks; instead, interest rate risk is an important driver of bond returns.

To test whether returns on longer-duration bonds respond more strongly to a decline in interest rate risk, we regress bond returns of varying durations on changes in interest rate risk. The empirical

specification is given by

$$\text{Return}_{n,t} = \beta_1 \Delta \text{UNC}_t + \beta_2 \text{Duration}_{n,t} + \beta_3 \Delta \text{UNC}_t \times \text{Duration}_{n,t} + \gamma X_{n,t} + \epsilon_{n,t}, \quad (10)$$

where $\text{Return}_{n,t}$ is the return on the n -year bond at the t -th FOMC, expressed in basis points, ΔUNC_t is a change in interest rate risk or principal component described in Section 3.1, and $\text{Duration}_{n,t}$ equals n , the bond's maturity (in years). We use bonds with maturities ranging from one year to twenty-nine years, in one-year increments.

The control variables $X_{n,t}$ include daily changes in the VIX, monetary policy shocks (measured by high-frequency changes in federal funds futures; see Nakamura and Steinsson (2018)), and their interactions with duration. These controls account for the possibility that changes in interest rate risk may be correlated with changes in aggregate uncertainty or the first moment of monetary policy, both of which could influence the sensitivity of returns across durations. The inclusion of the VIX reflects its widespread use as a proxy for uncertainty in the literature.⁶

Table 3 presents the results. Column (1) excludes control variables, while column (2) includes them. The coefficient of interest, β_3 , is estimated to be -0.42 in column (1) and -0.42 in column (2). The negative coefficient on the interaction term indicates that a 1% increase in interest rate risk reduces the return of bonds by 0.42 basis points more for each additional year of duration.

We also conduct sensitivity analyses using the principal component of several interest-rate-risk measures. Columns (3) and (4) show that the estimate of β_3 is negative and statistically significant. Additional robustness checks are reported in Appendix G. We conduct subsample analyses for the pre-crisis and post-crisis periods.

Discussion: relation to Hillenbrand (2025). Hillenbrand (2025) shows that a narrow window around FOMC announcements accounts for a substantial share of the secular decline in long-term Treasury yields. At first glance, it may be difficult to reconcile these findings with the proposed mechanism. If announcement returns are driven by a decline in uncertainty at FOMC meetings, why are they associated with a long-run secular decline in yields? In this section, we explore whether our mechanism and the findings of Hillenbrand (2025) can be reconciled through a complementary interpretation.

Figure 5 shows whether changes in interest rate uncertainty on announcement days are reversed after the announcement. We focus on FOMC days on which the change in our uncertainty measure falls below the 25th percentile of all announcement-day changes (“Bottom quartile”) or rises above

⁶Our option-implied measure captures market-implied uncertainty about future short-term interest rates, but does not separately identify risk-premium movements. We control for changes in the VIX and their interactions with duration to partly account for broader market uncertainty.

Table 3: The Sensitivity of Returns to Changes in Uncertainty by Duration.

	(1)	(2)	(3)	(4)
IRU	-4.61*** (1.18)	-2.65*** (0.95)		
Duration $\times$ IRU	-0.42** (0.18)	-0.42** (0.17)		
Prin. Comp.			-9.51*** (2.01)	-4.85** (1.99)
Duration $\times$ Prin. Comp.			-0.78** (0.38)	-0.77** (0.36)
MP shock		-255.25*** (46.02)		-259.55*** (47.72)
Duration $\times$ MP shock		-11.76 (9.56)		-12.41 (9.74)
Controls		✓		✓
R ²	0.11	0.17	0.09	0.15
Observations	5,974	5,974	5,974	5,974

Note: Table 3 reports results from pooled regressions of Treasury bond returns, measured in basis points, from 1994 to 2019. The main explanatory variables are changes in uncertainty, duration, and their interaction. Columns (1) and (2) use interest rate risk, while columns (3) and (4) use the principal component (PC) as the uncertainty measure. Control variables include MPS, the VIX, and their interactions with duration. Standard errors, clustered by time, are reported in parentheses. The regression equation is

$$\text{Return}_{n,t} = \beta_1 \Delta \text{UNC}_t + \beta_2 \text{Duration}_{n,t} + \beta_3 \Delta \text{UNC}_t \times \text{Duration}_{n,t} + \gamma X_{n,t} + \epsilon_{n,t},$$

where $\text{Return}_{n,t}$ is the return on bonds with duration n years at the t -th FOMC, ΔUNC_t is the change in uncertainty at the t -th FOMC, and $\text{Duration}_{n,t}$ equals n , where bonds mature in n years. Significance levels: *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$.

the 75th percentile (“Top quartile”). We plot the cumulative change in uncertainty from 30 trading days before the event to 90 trading days after the event. The figure shows that when uncertainty falls sharply on the FOMC day, the subsequent rebound is only partial. Interestingly, when the interest rate uncertainty increases, the cumulative change is reversed soon afterward. Thus, monetary policy announcements have asymmetric effects on interest rate uncertainty.

If announcement-induced declines in uncertainty are not fully transitory, their effects on yields may persist. Thus, the uncertainty-resolution mechanism is consistent with the possibility that narrow FOMC announcement windows contribute to a long-run decline in Treasury yields. However, this evidence should be interpreted cautiously, as we document persistence only over a 90-trading-day horizon.⁷ Nonetheless, this evidence complements the findings of [Hillenbrand \(2025\)](#).

⁷Extending the analysis to longer horizons would make identification substantially less clean, as other macroeco-

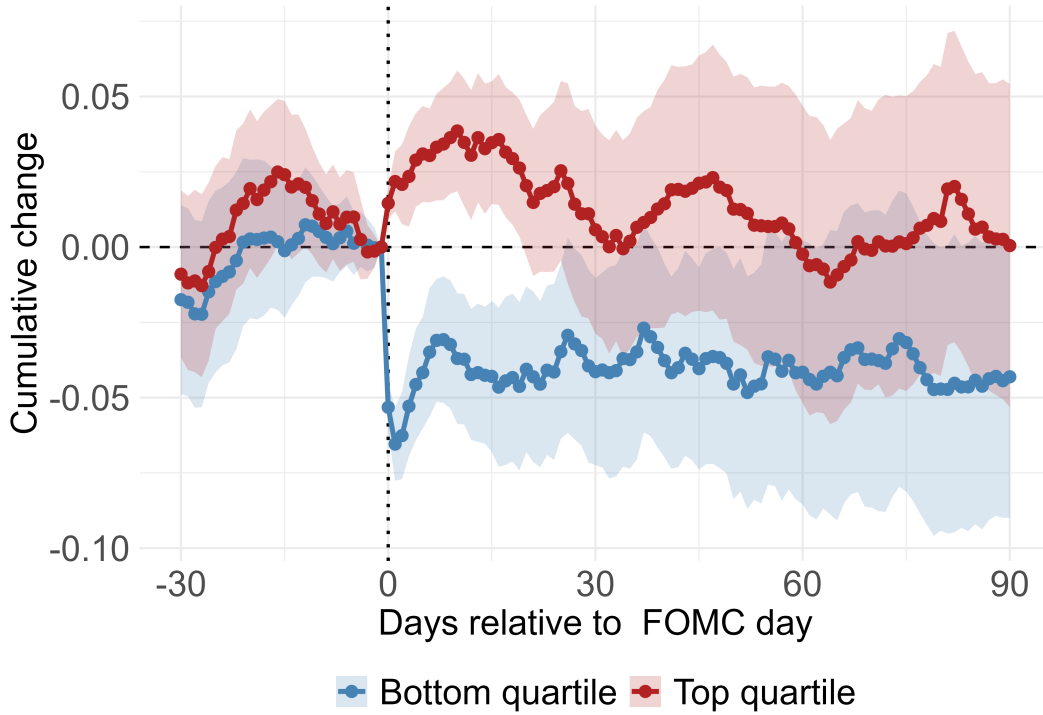


Figure 5: Persistence of Cumulative Uncertainty Changes after FOMC Announcements

Note: This figure plots the cumulative change in interest rate uncertainty around FOMC days on which the change in interest rate uncertainty falls below the 25th percentile of all announcements (“Bottom quartile”) or rises above the 75th percentile (“Top quartile”). Day 0 denotes the FOMC announcement day. The solid line shows the average cumulative change in IRU, and the shaded area represents Newey–West two-standard-error bands.

5 Empirical Analysis of Equities

5.1 Average Return on FOMC Days

We empirically examine whether short- or long-duration equities exhibit higher returns on announcement days. Equities are sorted into ten deciles based on cash-flow duration from the previous quarter, with portfolios rebalanced quarterly. Figure 6(a) presents the average portfolio returns in basis points with 95% confidence intervals based on Newey–West standard errors, as a function of cash-flow duration. The results indicate that the average returns on short- and long-duration equities are statistically indistinguishable. Specifically, the shortest-duration portfolio has an average return of 28.9 basis points, the second-shortest-duration portfolio has 24.4 basis points, and the longest-duration portfolio has 21.0 basis points. This empirical pattern differs from that of Treasury bonds, for which long-duration bonds exhibit higher returns.

economic and financial shocks are more likely to confound uncertainty dynamics.

We present results using alternative measures of duration. Figure 6(b) uses the book-to-market ratio as a proxy for duration, following Lettau and Wachter (2007) and Hansen, Heaton, and Li (2008). The results indicate that returns on low- and high-book-to-market ratio portfolios are not statistically distinguishable. Figure 6(c) uses the Gormsen and Lazarus (2023) measure of duration. This measure indicates that the longest-duration portfolio, Portfolio 1, earns a higher average return.

Importantly, the term structure of average returns on FOMC days contrasts sharply with that of average monthly returns. Figure 7(a) confirms the findings of Weber (2018); monthly average returns for portfolios sorted by cash-flow duration decline with duration. The shortest-duration portfolio earns 11.4 basis points, whereas the longest-duration portfolio earns only 0.3 basis points.⁸

The other duration measures show similar patterns. Figure 7(b) shows monthly average returns for portfolios sorted by book-to-market ratio, again displaying the downward-sloping term structure documented by Lettau and Wachter (2007). The shortest-duration portfolio earns 10.6 basis points and the longest-duration portfolio earns 2.5 basis points. Because the book-to-market ratio is negatively correlated with duration, this result is consistent with the cash-flow-duration sorting. Figure 7(c), based on Gormsen and Lazarus (2023), also shows that short-duration portfolios earn higher average monthly returns.

Although the term structure of returns on FOMC days varies across duration measures, a consistent pattern emerges: long-duration equities perform better relative to short-duration equities on FOMC days than they do in monthly data. This finding is consistent with the theoretical determinants of return differences in equation (3). The equation implies that, under certain conditions, the larger decline in interest rate risk raises returns on long-duration equities more than on short-duration equities. Because market-implied interest-rate uncertainty tends to decline more on FOMC days, the relative performance of long-duration equities improves on these days compared with the pattern in average monthly returns.

We estimate portfolio-level excess returns using risk factor regressions. The empirical specification is as follows:

$$y_t^m = \alpha^m + \beta_s^m X_{s,t} + \epsilon_t^m, \quad (11)$$

where y_t^m denotes the excess return of portfolio m at time t , with $m \in \{1, \dots, 10\}$ indexing portfolios sorted by duration. α^m is the model-specific pricing error, and β_s^m is the factor loading on the risk factors $X_{s,t}$. The risk factors are the Fama–French three factors.

Panel A of Table 4 reports mean excess returns (OLS estimates) and pricing errors from the three-factor model. The results do not reveal a clear upward or downward term structure. The three-factor model provides limited explanatory power for returns on FOMC days because the

⁸Monthly returns are computed as the price difference between the beginning and end of the month, converted to daily returns by dividing by the number of business days. Portfolios are formed by sorting equities into ten deciles based on duration from the previous quarter.

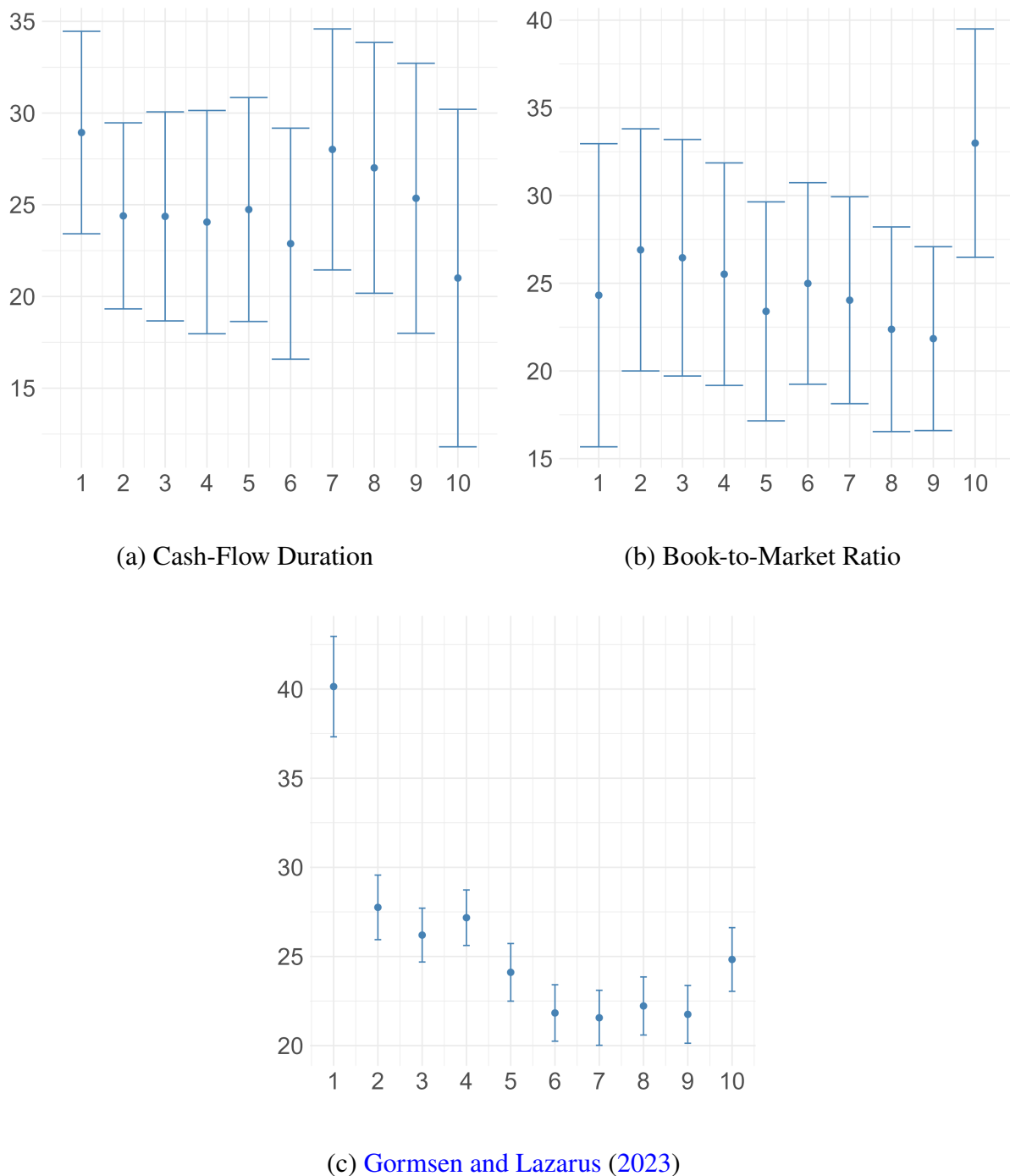


Figure 6: Average Returns Conditional on Duration.

Note: The horizontal axis represents portfolio rank. The vertical axis shows the average return for each portfolio. Panel (a) presents portfolios sorted on cash-flow duration from short (1) to long (10) as defined by [Weber \(2018\)](#), Panel (b) uses the book-to-market ratio from low (1, considered as long) to high (10, considered as short), and Panel (c) is based on [Gormsen and Lazarus \(2023\)](#) from long (1) to short (10). Confidence intervals are based on Newey–West standard errors.

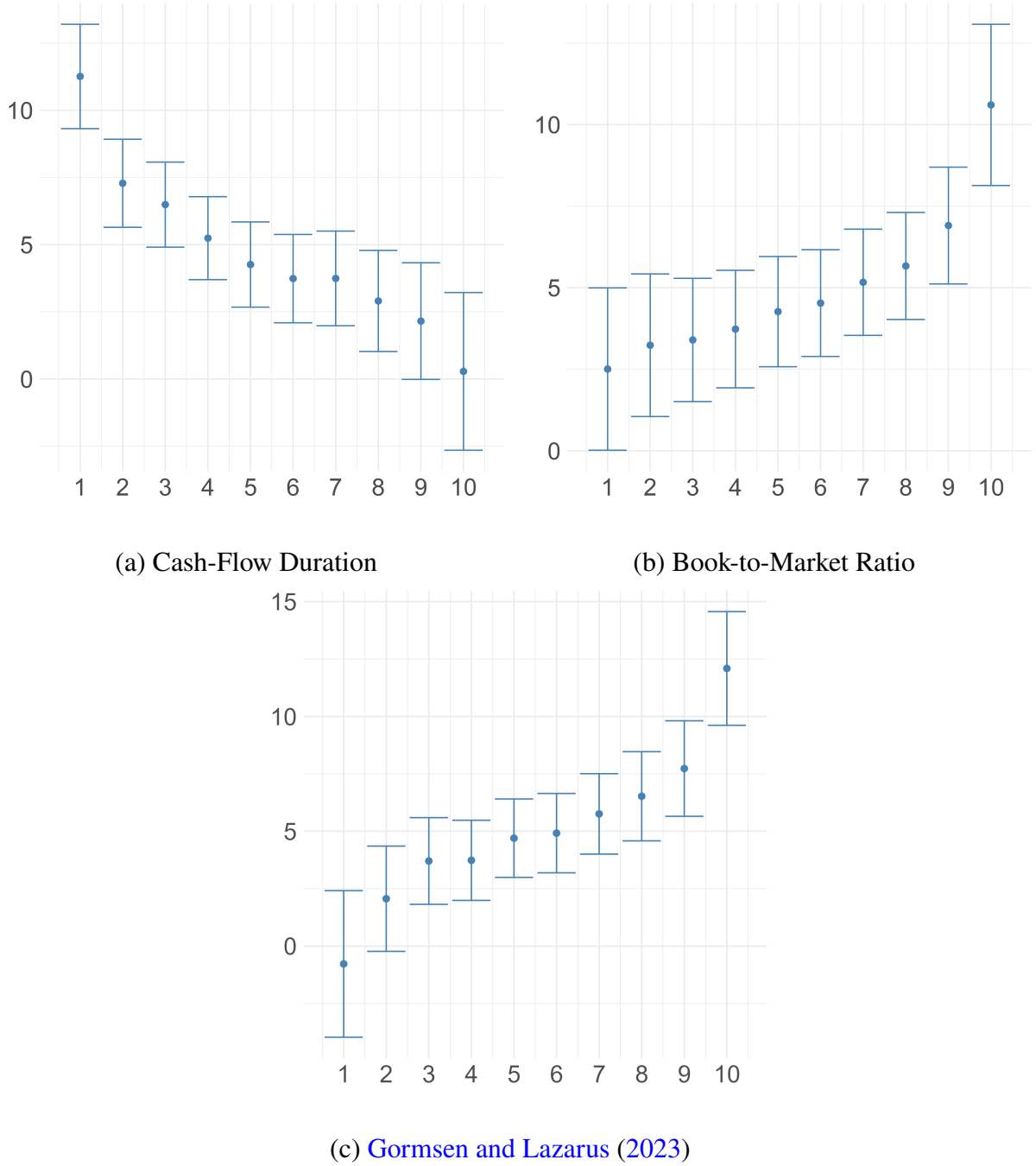


Figure 7: Monthly Average Returns Conditional on Duration.

Note: The horizontal axis represents portfolio rank. The vertical axis represents the average monthly return for each portfolio, expressed in basis points. Returns are converted from monthly to daily returns by dividing the monthly returns by the number of business days in the month. Panel (a) presents portfolios sorted on cash-flow duration from short (1) to long (10) as defined by [Weber \(2018\)](#), Panel (b) uses the book-to-market ratio from low (1, considered as long) to high (10, considered as short), and Panel (c) is based on [Gormsen and Lazarus \(2023\)](#) from long (1) to short (10).

Table 4: Average Returns of Portfolios Sorted by Duration.

Portfolio	1	2	3	4	5	6	7	8	9	10
Panel A: FOMC days										
Average returns	28.9 (2.8)	24.4 (2.6)	24.4 (2.9)	24.1 (3.1)	24.7 (3.1)	22.9 (3.2)	28.0 (3.3)	27.0 (3.5)	25.4 (3.8)	21.0 (4.7)
CAPM α	28.2 (2.8)	23.4 (2.6)	23.2 (3.0)	22.6 (3.1)	22.7 (3.2)	20.3 (3.2)	25.1 (3.3)	24.1 (3.5)	21.3 (3.8)	15.5 (3.6)
CAPM β	0.19 (0.14)	0.26 (0.15)	0.32 (0.16)	0.38 (0.17)	0.53 (0.17)	0.67 (0.17)	0.76 (0.18)	0.76 (0.17)	1.02 (0.17)	1.33 (0.32)
FF3 α	28.4 (2.8)	22.2 (2.6)	23.0 (2.9)	22.7 (3.1)	22.5 (3.1)	21.2 (3.1)	26.0 (3.3)	25.1 (3.4)	22.7 (3.7)	17.4 (3.2)
Panel B: Average Monthly Returns										
Average returns	11.4 (1.0)	7.2 (0.8)	6.4 (0.8)	5.2 (0.8)	4.3 (0.8)	3.7 (0.8)	3.8 (0.9)	2.9 (1.0)	2.2 (1.1)	0.3 (1.5)
CAPM α	10.5 (1.0)	6.4 (0.8)	5.6 (0.8)	4.4 (0.7)	3.4 (0.7)	2.7 (0.8)	2.6 (0.8)	1.6 (0.9)	0.5 (1.0)	-1.7 (1.3)
CAPM β	1.4 (0.25)	1.4 (0.23)	1.3 (0.22)	1.3 (0.21)	1.5 (0.21)	1.6 (0.21)	1.8 (0.23)	2.1 (0.23)	2.6 (0.27)	3.0 (0.37)
FF3 α	9.4 (1.0)	5.3 (0.8)	4.5 (0.8)	3.4 (0.8)	2.5 (0.8)	1.9 (0.8)	2.0 (0.9)	1.3 (0.9)	0.6 (1.0)	-1.2 (1.4)
Panel C: Properties of Portfolios										
Duration	9.5	14.3	16.1	17.3	18.2	19.1	19.9	20.8	21.7	23.6

Note: The table shows the time-series average of the portfolio returns. “Average returns” refers to the average returns on the portfolios, expressed in basis points. “FF3 α ” reports alphas from the three-factor model of [Fama and French \(1993\)](#). “1”–“10” represent the portfolios ordered from short to long duration. Panel A uses daily returns on FOMC days. Panel B uses monthly portfolio returns. In Panel B, returns are converted to a daily frequency by dividing monthly returns by the number of business days in the month. Panel C shows the average cash-flow duration of the portfolios in years. Standard errors in parentheses are Newey–West standard errors.

average returns and pricing errors are similar across portfolios. Panel B of Table 4 reports average monthly returns and pricing errors of the three-factor model. The results show that the average monthly returns decrease monotonically with duration.

5.2 Interest Rate Risk and Contemporaneous Regression

This section estimates the sensitivity of returns to changes in interest rate risk across durations. We separately estimate the following regression for each portfolio to examine the sensitivity of returns to changes in interest rate risk:

$$y_t^m = \alpha^m + \beta_{IRU}^m \Delta IRU_t + \epsilon_t^m,$$

where y_t^m is the return of portfolio m at time t expressed in basis points. Portfolios are indexed by $m \in \{1, \dots, 10\}$ and are constructed by sorting stocks into ten deciles based on cash-flow duration in the previous quarter.

Figure 8(a) presents the estimated coefficients β_{IRU}^m plotted against portfolio duration m , along with the 95% confidence intervals based on OLS standard errors. The coefficients decrease monotonically with duration: for the shortest-duration portfolio, the estimate is -9, while for the longest-duration portfolio, it is -18. This indicates that the return sensitivity to interest rate risk increases in absolute value with duration.

Figure 8(b) presents the estimated coefficients β_{IRU}^m for portfolios sorted by book-to-market ratio. The coefficients increase monotonically with the book-to-market ratio: the lowest book-to-market portfolio has an estimate of -16, while the highest has -11. Figure 8(c) presents the results based on Gormsen and Lazarus (2023). The long-duration portfolio also shows a larger exposure to interest rate risk. Since Figure 8(c) shows some nonmonotonicity, a formal statistical test is reported in Table 10 in Appendix H.

We estimate the following regression to assess whether returns on longer-duration equities are more sensitive to changes in interest rate risk:

$$\text{Return}_{i,t} = \beta_1 \text{Duration}_{i,t} + \beta_2 \Delta \text{IRU}_t \times \text{Duration}_{i,t} + \gamma X_{i,t} + \epsilon_{i,t}, \quad (12)$$

where i indexes firms and t indexes FOMC announcements, $\text{Return}_{i,t}$ denotes the stock return of firm i on the t -th FOMC day (in basis points), ΔIRU_t is the percent change in interest rate risk associated with the t -th FOMC, and $\text{Duration}_{i,t}$ is the duration of firm i at time t . $X_{i,t}$ denotes a vector of control variables, including firm and time fixed effects, firm characteristics such as size, profitability, leverage, and sales growth, the monetary policy surprise interacted with duration, and interactions between these characteristics and ΔIRU_t . All balance sheet variables are measured using information from the previous quarter. Robust standard errors are clustered at the firm and time levels.

Table 5 presents the regression results. Columns (1) and (2) use the baseline IRU measure and columns (3) and (4) use the principal component measure. Columns (2) and (4) include the controls. The coefficient of interest is negative and statistically significant across all the specifications. Thus, the sensitivity of returns with respect to interest rate risk increases in absolute value with duration.

Robustness check. Appendix E shows the placebo tests that randomly assign a change in interest rate risk from the entire distribution. Appendix H reports the sensitivity analyses across subsamples, alternative duration measures, and parameter values used to construct duration.

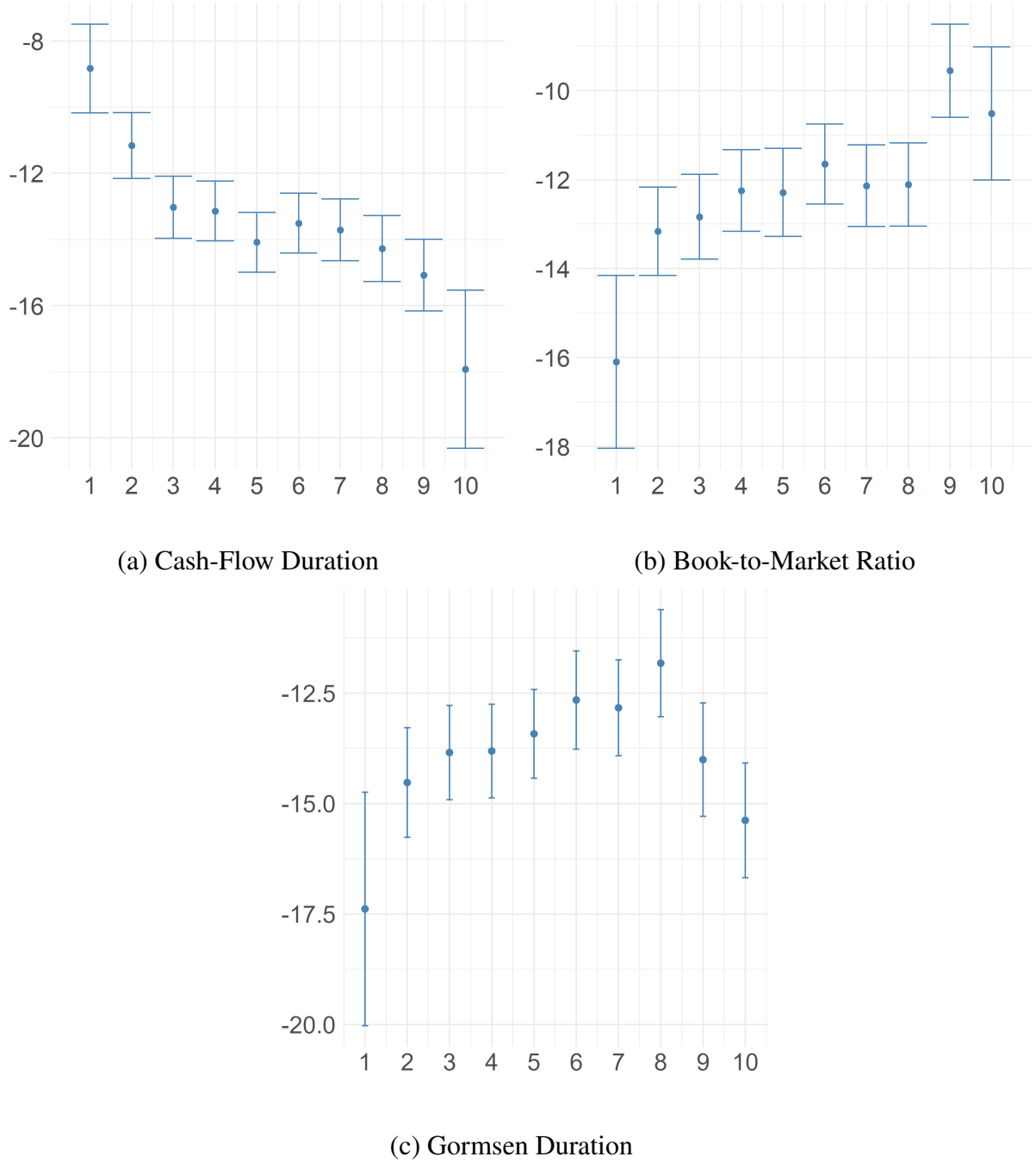


Figure 8: The Sensitivity of Returns to Interest Rate Risk Conditional on Duration.

Note: We separately regress portfolio returns on changes in the uncertainty measure: $y_t^m = \alpha^m + \beta^m \Delta IRU_t + \epsilon_t^m$, where y_t^m is the return of portfolio m at time t expressed in basis points, and ΔIRU_t is the percent change in interest rate risk. Portfolios are indexed by $m \in \{1, \dots, 10\}$. Panel (a) presents portfolios sorted on cash-flow duration from short (1) to long (10) as defined by [Weber \(2018\)](#), Panel (b) uses the book-to-market ratio from low (1, considered as long) to high (10, considered as short), and Panel (c) is based on [Gormsen and Lazarus \(2023\)](#) from long (1) to short (10). The figure plots the coefficients β^m for each portfolio m , along with their two-standard-error bands.

Table 5: The Sensitivity of Returns to Interest Rate Risk Conditional on Duration.

	(1)	(2)	(3)	(4)
IRU $\times$ Duration	-0.55*** (0.17)	-0.61*** (0.18)		
Prin. Comp. $\times$ Duration			-1.22*** (0.35)	-1.35*** (0.39)
MP shock $\times$ Duration		-4.30 (8.01)		-3.46 (8.60)
R ²	0.07	0.07	0.07	0.07
Observations	686,324	623,073	686,324	623,073

Note: This table reports estimates from pooled regressions of equity returns around FOMC announcements. The dependent variable is the equity return, measured in basis points. The key regressors are the change in interest rate uncertainty, firm duration, and their interaction. Columns (1) and (2) use the change in interest rate uncertainty. Columns (3) and (4) use the first principal component of interest rate uncertainty measures. Columns (2) and (4) include firm-level controls and interactions between the controls and the corresponding uncertainty measure. Firm-level controls are size, profitability, leverage, and sales growth. Standard errors, clustered at the firm and time levels, are reported in parentheses. Significance levels are denoted by *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$.

6 Conclusion

This paper provides empirical evidence and a theoretical framework to understand cross-sectional asset returns on FOMC announcement days. Specifically, it investigates the types of uncertainty resolved by FOMC announcements and the factors underlying heterogeneity in stock returns. These findings are consistent with the literature emphasizing risk-based explanations of the elevated excess returns observed on announcement days. To evaluate the importance of interest rate risk, this paper develops a theory relating announcement returns to asset duration and presents supporting empirical evidence.

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Appendix A Proofs

A.1 General Solutions

This appendix derives the expected announcement returns reported in Section 2. Throughout the appendix, an overbar denotes an unconditional mean, and a hat denotes a log deviation for the primitive variables.

For compact notation, let

$$H(s) \equiv \left[C_2^{1-\frac{1}{\psi}} + \beta(s)C_3(s)^{1-\frac{1}{\psi}} + \beta(s)^2C_4(s)^{1-\frac{1}{\psi}} \right]^{\frac{\frac{1}{\psi}-\gamma}{1-\frac{1}{\psi}}}.$$

Define

$$\bar{\beta} \equiv E[\beta(s)], \quad \bar{C}_j \equiv E[C_j(s)], \quad \bar{X}_j \equiv E[X_j(s)], \quad j \in \{3, 4\},$$

and

$$\hat{\beta}(s) \equiv \log \beta(s) - \log \bar{\beta}, \quad \hat{c}_j(s) \equiv \log C_j(s) - \log \bar{C}_j, \quad \hat{x}_j(s) \equiv \log X_j(s) - \log \bar{X}_j.$$

The shares of periods 3 and 4 in the intertemporal utility aggregator, evaluated at the approximation point, are

$$a_3 \equiv \frac{\bar{\beta}\bar{C}_3^{1-\frac{1}{\psi}}}{C_2^{1-\frac{1}{\psi}} + \bar{\beta}\bar{C}_3^{1-\frac{1}{\psi}} + \bar{\beta}^2\bar{C}_4^{1-\frac{1}{\psi}}}, \quad a_4 \equiv \frac{\bar{\beta}^2\bar{C}_4^{1-\frac{1}{\psi}}}{C_2^{1-\frac{1}{\psi}} + \bar{\beta}\bar{C}_3^{1-\frac{1}{\psi}} + \bar{\beta}^2\bar{C}_4^{1-\frac{1}{\psi}}}.$$

Let

$$\bar{H} \equiv \left[C_2^{1-\frac{1}{\psi}} + \bar{\beta}\bar{C}_3^{1-\frac{1}{\psi}} + \bar{\beta}^2\bar{C}_4^{1-\frac{1}{\psi}} \right]^{\frac{\frac{1}{\psi}-\gamma}{1-\frac{1}{\psi}}}.$$

A first-order log-linear approximation of $H(s)$ around $(C_2, \bar{C}_3, \bar{C}_4, \bar{\beta})$ gives

$$\frac{H(s)}{\bar{H}} \simeq 1 + \left(\frac{1}{\psi} - \gamma \right) \left[\frac{a_3 + 2a_4}{1 - \frac{1}{\psi}} \hat{\beta}(s) + a_3 \hat{c}_3(s) + a_4 \hat{c}_4(s) \right]. \quad (13)$$

The coefficient $a_3 + 2a_4$ appears because $\beta(s)$ enters once in the period-3 component of the utility

aggregator and twice in the period-4 component.

Define the first-order state-dependent component of the announcement valuation term as

$$\widehat{H}(s) \equiv \frac{a_3 + 2a_4}{1 - \frac{1}{\psi}} \widehat{\beta}(s) + a_3 \widehat{c}_3(s) + a_4 \widehat{c}_4(s). \quad (14)$$

Then (13) can be written compactly as

$$\frac{H(s)}{\bar{H}} \simeq 1 + \left(\frac{1}{\psi} - \gamma \right) \widehat{H}(s). \quad (15)$$

Post-announcement prices. After the announcement reveals state s , the investor trades the short- and long-duration assets in period 2. We assume that equilibrium period-2 consumption is identical across states, $C_2(s) = C_2$, while the period-2 marginal conditions hold state by state. Up to a common positive factor, marginal utilities are

$$U_{C_2}(s) \propto H(s) C_2^{-\frac{1}{\psi}}, \quad U_{C_3}(s) \propto H(s) \beta(s) C_3(s)^{-\frac{1}{\psi}},$$

and

$$U_{C_4}(s) \propto H(s) \beta(s)^2 C_4(s)^{-\frac{1}{\psi}}.$$

The period-2 first-order conditions therefore imply

$$P_2^S(s) = \beta(s) X_3(s) \left(\frac{C_3(s)}{C_2} \right)^{-\frac{1}{\psi}}, \quad (16)$$

$$P_2^L(s) = \beta(s)^2 X_4(s) \left(\frac{C_4(s)}{C_2} \right)^{-\frac{1}{\psi}}. \quad (17)$$

Define the prices at the approximation point as

$$\bar{P}_2^S \equiv \bar{\beta} \bar{X}_3 \left(\frac{\bar{C}_3}{C_2} \right)^{-\frac{1}{\psi}}, \quad \bar{P}_2^L \equiv \bar{\beta}^2 \bar{X}_4 \left(\frac{\bar{C}_4}{C_2} \right)^{-\frac{1}{\psi}}.$$

Their first-order approximations are

$$\frac{P_2^S(s)}{\bar{P}_2^S} \simeq 1 + \widehat{\beta}(s) + \widehat{x}_3(s) - \frac{1}{\psi} \widehat{c}_3(s), \quad (18)$$

$$\frac{P_2^L(s)}{\bar{P}_2^L} \simeq 1 + 2\hat{\beta}(s) + \hat{x}_4(s) - \frac{1}{\psi}\hat{c}_4(s). \quad (19)$$

For further compactness, define

$$\hat{p}_2^S(s) \equiv \hat{\beta}(s) + \hat{x}_3(s) - \frac{1}{\psi}\hat{c}_3(s),$$

and

$$\hat{p}_2^L(s) \equiv 2\hat{\beta}(s) + \hat{x}_4(s) - \frac{1}{\psi}\hat{c}_4(s).$$

Pre-announcement prices and announcement returns. Before the announcement, the state is unknown. The period-1 first-order conditions imply

$$P_1^S = \frac{E[U_{C_3}(s)X_3(s)]}{E[U_{C_2}(s)]}, \quad P_1^L = \frac{E[U_{C_4}(s)X_4(s)]}{E[U_{C_2}(s)]}.$$

Using the post-announcement pricing equations,

$$P_1^S = \frac{E[H(s)P_2^S(s)]}{E[H(s)]}, \quad (20)$$

$$P_1^L = \frac{E[H(s)P_2^L(s)]}{E[H(s)]}. \quad (21)$$

The expected announcement returns are therefore

$$R^S \equiv E_1 \left[\frac{P_2^S(s)}{P_1^S} \right] = \frac{E[H(s)]E[P_2^S(s)]}{E[H(s)P_2^S(s)]}, \quad (22)$$

$$R^L \equiv E_1 \left[\frac{P_2^L(s)}{P_1^L} \right] = \frac{E[H(s)]E[P_2^L(s)]}{E[H(s)P_2^L(s)]}. \quad (23)$$

Using

$$E[H(s)P_2^j(s)] = E[H(s)]E[P_2^j(s)] + \text{Cov}(H(s), P_2^j(s)), \quad j \in \{S, L\},$$

together with (15), (18), and (19), gives

$$\frac{\text{Cov}(H(s), P_2^S(s))}{E[H(s)]E[P_2^S(s)]} \simeq \left(\frac{1}{\psi} - \gamma \right) \text{Cov}(\hat{H}, \hat{p}_2^S), \quad (24)$$

$$\frac{\text{Cov}(H(s), P_2^L(s))}{E[H(s)]E[P_2^L(s)]} \simeq \left(\frac{1}{\psi} - \gamma\right) \text{Cov}\left(\widehat{H}, \widehat{p}_2^L\right). \quad (25)$$

Using $1/(1+z) \simeq 1-z$, the expected announcement returns are therefore

$$R^S \simeq 1 + \left(\gamma - \frac{1}{\psi}\right) \text{Cov}\left(\widehat{H}, \widehat{\beta} + \widehat{x}_3 - \frac{1}{\psi}\widehat{c}_3\right), \quad (26)$$

$$R^L \simeq 1 + \left(\gamma - \frac{1}{\psi}\right) \text{Cov}\left(\widehat{H}, 2\widehat{\beta} + \widehat{x}_4 - \frac{1}{\psi}\widehat{c}_4\right). \quad (27)$$

Subtracting (26) from (27) gives

$$R^L - R^S \simeq \left(\gamma - \frac{1}{\psi}\right) \text{Cov}\left(\widehat{H}, \widehat{\beta} - \widehat{x}_3 + \widehat{x}_4 + \frac{1}{\psi}(\widehat{c}_3 - \widehat{c}_4)\right). \quad (28)$$

Equation (28) shows that the long-minus-short announcement return depends on the covariance between the announcement valuation term and the additional exposure of the long-duration asset. The latter consists of discount-factor exposure, differences in short- and long-horizon cash-flow news, and differences in marginal-utility exposure across horizons.

Bonds. For bonds, cash flows are deterministic, so

$$\widehat{x}_3(s) = \widehat{x}_4(s) = 0.$$

Equations (26) and (27) reduce to

$$R^S \simeq 1 + \left(\gamma - \frac{1}{\psi}\right) \text{Cov}\left(\widehat{H}, \widehat{\beta} - \frac{1}{\psi}\widehat{c}_3\right), \quad (29)$$

$$R^L \simeq 1 + \left(\gamma - \frac{1}{\psi}\right) \text{Cov}\left(\widehat{H}, 2\widehat{\beta} - \frac{1}{\psi}\widehat{c}_4\right). \quad (30)$$

Hence, the long-minus-short announcement return is

$$R^L - R^S \simeq \left(\gamma - \frac{1}{\psi}\right) \text{Cov}\left(\widehat{H}, \widehat{\beta} + \frac{1}{\psi}(\widehat{c}_3 - \widehat{c}_4)\right). \quad (31)$$

A.2 Proof of Proposition 1

Proof. Using

$$\widehat{H} = \frac{a_3 + 2a_4}{1 - \frac{1}{\psi}} \widehat{\beta} + a_3 \widehat{c}_3 + a_4 \widehat{c}_4,$$

the discount-factor variance term in the short-duration bond premium is

$$\left(\gamma - \frac{1}{\psi} \right) \frac{a_3 + 2a_4}{1 - \frac{1}{\psi}} \text{Var}(\widehat{\beta}),$$

whereas that in the long-duration bond premium is

$$2 \left(\gamma - \frac{1}{\psi} \right) \frac{a_3 + 2a_4}{1 - \frac{1}{\psi}} \text{Var}(\widehat{\beta}).$$

Thus, the contribution of discount-factor risk to the announcement premium is exactly twice as large for the long-duration bond. In this sense, the long-duration bond is more exposed to the discount-factor risk resolved at the announcement.

To establish the return ranking when consumption risk is small, parameterize discount-factor and consumption risk as

$$\widehat{\beta} = \sigma_\beta z_\beta, \quad \widehat{c}_j = \varepsilon \sigma_\beta \widetilde{c}_j, \quad j \in \{3, 4\},$$

where

$$\text{Var}(z_\beta) = 1, \quad \sigma_\beta > 0,$$

and $\widetilde{c}_3$ and $\widetilde{c}_4$ have finite second moments. As ε varies, hold fixed the joint distribution of $(z_\beta, \widetilde{c}_3, \widetilde{c}_4)$. The parameter ε therefore measures the magnitude of consumption risk relative to discount-factor risk.

Then

$$\widehat{H} = \sigma_\beta \left[\frac{a_3 + 2a_4}{1 - \frac{1}{\psi}} z_\beta + \varepsilon (a_3 \widetilde{c}_3 + a_4 \widetilde{c}_4) \right].$$

Equations (29) and (30) imply

$$R^S - 1 \simeq \left(\gamma - \frac{1}{\psi} \right) \sigma_\beta^2 \left[\frac{a_3 + 2a_4}{1 - \frac{1}{\psi}} + O(\varepsilon) \right], \quad (32)$$

$$R^L - 1 \simeq \left(\gamma - \frac{1}{\psi} \right) \sigma_\beta^2 \left[2 \frac{a_3 + 2a_4}{1 - \frac{1}{\psi}} + O(\varepsilon) \right], \quad (33)$$

and hence

$$R^L - R^S \simeq \left(\gamma - \frac{1}{\psi} \right) \sigma_\beta^2 \left[\frac{a_3 + 2a_4}{1 - \frac{1}{\psi}} + O(\varepsilon) \right]. \quad (34)$$

At $\varepsilon = 0$,

$$R^S - 1 \simeq \left(\gamma - \frac{1}{\psi} \right) \sigma_\beta^2 \frac{a_3 + 2a_4}{1 - \frac{1}{\psi}}, \quad (35)$$

$$R^L - 1 \simeq 2 \left(\gamma - \frac{1}{\psi} \right) \sigma_\beta^2 \frac{a_3 + 2a_4}{1 - \frac{1}{\psi}}, \quad (36)$$

$$R^L - R^S \simeq \left(\gamma - \frac{1}{\psi} \right) \sigma_\beta^2 \frac{a_3 + 2a_4}{1 - \frac{1}{\psi}}. \quad (37)$$

Since $a_3 > 0$ and $a_4 > 0$,

$$a_3 + 2a_4 > 0.$$

The maintained condition

$$\left(\gamma - \frac{1}{\psi} \right) \frac{1}{1 - \frac{1}{\psi}} > 0$$

therefore implies that all three expressions are strictly positive at $\varepsilon = 0$. By continuity, there exists $\bar{\varepsilon} > 0$ such that they remain strictly positive for all $0 \leq \varepsilon < \bar{\varepsilon}$. Hence,

$$R^L > R^S > 1.$$

□

A.3 Proof of Proposition 2

Proof. Write

$$\hat{\beta} = \sigma_\beta z_\beta, \quad \text{Var}(z_\beta) = 1,$$

and parameterize announcement-related consumption risk as

$$\widehat{c}_3 = \varepsilon \widetilde{c}_3, \quad \widehat{c}_4 = \varepsilon \widetilde{c}_4,$$

where $\widetilde{c}_3$ and $\widetilde{c}_4$ have finite second moments. When varying σ_β , we hold fixed the joint distribution of $(z_\beta, \widetilde{c}_3, \widetilde{c}_4)$ and the approximation point.

Using the definition of $\widehat{H}$,

$$\widehat{H} = \frac{a_3 + 2a_4}{1 - \frac{1}{\psi}} \sigma_\beta z_\beta + \varepsilon (a_3 \widetilde{c}_3 + a_4 \widetilde{c}_4).$$

Equations (29) and (30) imply

$$R^S - 1 \simeq \left(\gamma - \frac{1}{\psi} \right) \text{Cov} \left(\frac{a_3 + 2a_4}{1 - \frac{1}{\psi}} \sigma_\beta z_\beta + \varepsilon (a_3 \widetilde{c}_3 + a_4 \widetilde{c}_4), \sigma_\beta z_\beta - \frac{\varepsilon}{\psi} \widetilde{c}_3 \right), \quad (38)$$

$$R^L - 1 \simeq \left(\gamma - \frac{1}{\psi} \right) \text{Cov} \left(\frac{a_3 + 2a_4}{1 - \frac{1}{\psi}} \sigma_\beta z_\beta + \varepsilon (a_3 \widetilde{c}_3 + a_4 \widetilde{c}_4), 2\sigma_\beta z_\beta - \frac{\varepsilon}{\psi} \widetilde{c}_4 \right). \quad (39)$$

Differentiating with respect to σ_β gives

$$\begin{aligned} \frac{\partial R^S}{\partial \sigma_\beta} \simeq \left(\gamma - \frac{1}{\psi} \right) & \left[2 \frac{a_3 + 2a_4}{1 - \frac{1}{\psi}} \sigma_\beta \right. \\ & \left. + \varepsilon \left\{ a_3 \text{Cov}(\widetilde{c}_3, z_\beta) + a_4 \text{Cov}(\widetilde{c}_4, z_\beta) - \frac{a_3 + 2a_4}{\psi \left(1 - \frac{1}{\psi} \right)} \text{Cov}(z_\beta, \widetilde{c}_3) \right\} \right], \quad (40) \end{aligned}$$

and

$$\begin{aligned} \frac{\partial R^L}{\partial \sigma_\beta} \simeq \left(\gamma - \frac{1}{\psi} \right) & \left[4 \frac{a_3 + 2a_4}{1 - \frac{1}{\psi}} \sigma_\beta \right. \\ & \left. + \varepsilon \left\{ 2a_3 \text{Cov}(\widetilde{c}_3, z_\beta) + 2a_4 \text{Cov}(\widetilde{c}_4, z_\beta) - \frac{a_3 + 2a_4}{\psi \left(1 - \frac{1}{\psi} \right)} \text{Cov}(z_\beta, \widetilde{c}_4) \right\} \right]. \quad (41) \end{aligned}$$

Therefore,

$$\frac{\partial R^L}{\partial \sigma_\beta} - \frac{\partial R^S}{\partial \sigma_\beta} \simeq \left(\gamma - \frac{1}{\psi} \right) \left[2 \frac{a_3 + 2a_4}{1 - \frac{1}{\psi}} \sigma_\beta + \varepsilon \left\{ a_3 \text{Cov}(\tilde{c}_3, z_\beta) + a_4 \text{Cov}(\tilde{c}_4, z_\beta) + \frac{a_3 + 2a_4}{\psi \left(1 - \frac{1}{\psi} \right)} \text{Cov}(z_\beta, \tilde{c}_3 - \tilde{c}_4) \right\} \right]. \quad (42)$$

At $\varepsilon = 0$,

$$\frac{\partial R^L}{\partial \sigma_\beta} - \frac{\partial R^S}{\partial \sigma_\beta} \simeq 2 \left(\gamma - \frac{1}{\psi} \right) \frac{a_3 + 2a_4}{1 - \frac{1}{\psi}} \sigma_\beta.$$

Since $a_3 > 0$ and $a_4 > 0$,

$$a_3 + 2a_4 > 0.$$

Hence, the maintained condition

$$\left(\gamma - \frac{1}{\psi} \right) \frac{1}{1 - \frac{1}{\psi}} > 0$$

and $\sigma_\beta > 0$ imply

$$\frac{\partial R^L}{\partial \sigma_\beta} - \frac{\partial R^S}{\partial \sigma_\beta} > 0$$

at $\varepsilon = 0$.

Because the covariance terms in (42) are finite, the expression is continuous in ε . Therefore, there exists $\bar{\varepsilon} > 0$ such that, for all $0 \leq \varepsilon < \bar{\varepsilon}$,

$$\frac{\partial R^L}{\partial \sigma_\beta} > \frac{\partial R^S}{\partial \sigma_\beta}.$$

Thus, when announcement-related consumption risk is sufficiently small relative to discount-factor risk, resolving more discount-factor risk raises the announcement return of the long-duration bond more than that of the short-duration bond. $\square$

A.4 Proof of Proposition 3

Proof. Parameterize announcement-related consumption risk as

$$\hat{c}_3 = \varepsilon \tilde{c}_3, \quad \hat{c}_4 = \varepsilon \tilde{c}_4,$$

where ε measures the magnitude of consumption risk relative to discount-factor risk. Then

$$\widehat{H} = \frac{a_3 + 2a_4}{1 - \frac{1}{\psi}} \widehat{\beta} + \varepsilon (a_3 \widetilde{c}_3 + a_4 \widetilde{c}_4).$$

From equation (28),

$$R^L - R^S \simeq \left(\gamma - \frac{1}{\psi} \right) \text{Cov} \left(\frac{a_3 + 2a_4}{1 - \frac{1}{\psi}} \widehat{\beta} + \varepsilon (a_3 \widetilde{c}_3 + a_4 \widetilde{c}_4), \widehat{\beta} - \widehat{x}_3 + \widehat{x}_4 + \frac{\varepsilon}{\psi} (\widetilde{c}_3 - \widetilde{c}_4) \right).$$

Collecting the terms that remain as $\varepsilon \rightarrow 0$ gives

$$R^L - R^S \simeq \left(\gamma - \frac{1}{\psi} \right) \frac{a_3 + 2a_4}{1 - \frac{1}{\psi}} \left[\text{Var}(\widehat{\beta}) + \text{Cov}(\widehat{\beta}, \widehat{x}_4) - \text{Cov}(\widehat{\beta}, \widehat{x}_3) \right] + O(\varepsilon).$$

Thus, to first order in consumption risk, the long-short announcement return differential depends on discount-factor risk through $\text{Var}(\widehat{\beta})$ and on cash-flow news through its covariances with discount-factor news,

$$\text{Cov}(\widehat{\beta}, \widehat{x}_4) - \text{Cov}(\widehat{\beta}, \widehat{x}_3).$$

Because the model does not in general restrict these covariance terms, smooth consumption alone does not determine the sign of $R^L - R^S$. $\square$

A.5 Derivation of Equation 4

The long-short equity return differential is given by

$$R^L - R^S \simeq \left(\gamma - \frac{1}{\psi} \right) \text{Cov} \left(\widehat{H}, \widehat{\beta} - \widehat{x}_3 + \widehat{x}_4 + \frac{1}{\psi} (\widehat{c}_3 - \widehat{c}_4) \right). \quad (43)$$

Parameterize discount-factor risk as

$$\widehat{\beta} = \sigma_\beta z_\beta, \quad \text{Var}(z_\beta) = 1,$$

and consumption risk as

$$\widehat{c}_j = \varepsilon \sigma_\beta \widetilde{c}_j, \quad j \in \{3, 4\},$$

where ε measures the magnitude of consumption risk relative to discount-factor risk. Using

$$\widehat{H} = \frac{a_3 + 2a_4}{1 - \frac{1}{\psi}} \widehat{\beta} + a_3 \widehat{c}_3 + a_4 \widehat{c}_4,$$

equation (43) becomes

$$R^L - R^S \simeq \left(\gamma - \frac{1}{\psi} \right) \frac{a_3 + 2a_4}{1 - \frac{1}{\psi}} \left[\text{Var}(\widehat{\beta}) - \text{Cov}(\widehat{\beta}, \widehat{x}_3) + \text{Cov}(\widehat{\beta}, \widehat{x}_4) \right] + O(\varepsilon). \quad (44)$$

Let

$$\sigma_{x_j}^2 = \text{Var}(\widehat{x}_j), \quad \rho_{\beta x_j} = \text{Corr}(\widehat{\beta}, \widehat{x}_j).$$

Then

$$\text{Var}(\widehat{\beta}) = \sigma_{\beta}^2$$

and

$$\text{Cov}(\widehat{\beta}, \widehat{x}_j) = \rho_{\beta x_j} \sigma_{\beta} \sigma_{x_j}.$$

Holding fixed $\rho_{\beta x_j}$ and σ_{x_j} as σ_{β} varies, equation (44) can therefore be written as

$$R^L - R^S \simeq \left(\gamma - \frac{1}{\psi} \right) \frac{a_3 + 2a_4}{1 - \frac{1}{\psi}} \left[\sigma_{\beta}^2 - \rho_{\beta x_3} \sigma_{\beta} \sigma_{x_3} + \rho_{\beta x_4} \sigma_{\beta} \sigma_{x_4} \right] + O(\varepsilon). \quad (45)$$

Differentiating with respect to σ_{β} yields

$$\frac{\partial(R^L - R^S)}{\partial \sigma_{\beta}} \simeq \left(\gamma - \frac{1}{\psi} \right) \frac{a_3 + 2a_4}{1 - \frac{1}{\psi}} [2\sigma_{\beta} - \rho_{\beta x_3} \sigma_{x_3} + \rho_{\beta x_4} \sigma_{x_4}] + O(\varepsilon), \quad (46)$$

which is equation (4).

Appendix B Option-Implied Volatility

Risk-Neutral and Physical Measures

$$H(s) \equiv \left[C_2^{1-\frac{1}{\psi}} + \beta(s) C_3(s)^{1-\frac{1}{\psi}} + \beta(s)^2 C_4(s)^{1-\frac{1}{\psi}} \right]^{\xi},$$

where

$$\eta \equiv 1 - \frac{1}{\psi}, \quad \xi \equiv \frac{\frac{1}{\psi} - \gamma}{1 - \frac{1}{\psi}}.$$

Let

$$L(s) \equiv \frac{H(s)}{E^{\mathbb{P}}[H(s)]}.$$

Then $L(s)$ is the Radon-Nikodym derivative from the physical measure to the risk-neutral measure:

$$\frac{d\mathbb{Q}}{d\mathbb{P}}(s) = L(s).$$

The risk-neutral variance of $\hat{\beta}(s)$ is

$$\text{Var}^{\mathbb{Q}}(\hat{\beta}) = E^{\mathbb{Q}} \left[\hat{\beta}(s)^2 \right] - \left(E^{\mathbb{Q}} \left[\hat{\beta}(s) \right] \right)^2.$$

Using the change of measure,

$$E^{\mathbb{Q}} \left[\hat{\beta}(s) \right] = E^{\mathbb{P}} \left[L(s) \hat{\beta}(s) \right], \quad E^{\mathbb{Q}} \left[\hat{\beta}(s)^2 \right] = E^{\mathbb{P}} \left[L(s) \hat{\beta}(s)^2 \right].$$

Therefore,

$$\text{Var}^{\mathbb{Q}}(\hat{\beta}) = E^{\mathbb{P}} \left[L(s) \hat{\beta}(s)^2 \right] - \left(E^{\mathbb{P}} \left[L(s) \hat{\beta}(s) \right] \right)^2.$$

Equivalently,

$$\text{Var}^{\mathbb{Q}}(\hat{\beta}) = \text{Var}^{\mathbb{P}}(\hat{\beta}) + \text{Cov}^{\mathbb{P}} \left(L(s), \hat{\beta}(s)^2 \right) - 2E^{\mathbb{P}} \left[\hat{\beta}(s) \right] \text{Cov}^{\mathbb{P}} \left(L(s), \hat{\beta}(s) \right) - \left[\text{Cov}^{\mathbb{P}} \left(L(s), \hat{\beta}(s) \right) \right]^2.$$

Thus, risk-neutral variance equals physical variance plus a pricing-kernel weighting adjustment. If

$$E^{\mathbb{P}} \left[\hat{\beta}(s) \right] \simeq 0,$$

this becomes

$$\text{Var}^{\mathbb{Q}}(\hat{\beta}) \simeq \text{Var}^{\mathbb{P}}(\hat{\beta}) + \text{Cov}^{\mathbb{P}} \left(L(s), \hat{\beta}(s)^2 \right) - \left[\text{Cov}^{\mathbb{P}} \left(L(s), \hat{\beta}(s) \right) \right]^2.$$

This expression shows that the difference between physical and risk-neutral variance depends on how discount-factor volatility covaries with the pricing kernel.

Mapping to Option-Implied Volatility. Suppose that the one-period pricing relation satisfies

$$\beta(s)(1 + i(s)) = 1,$$

where $i(s)$ is the net risk-free interest rate. Then

$$\beta(s) = \frac{1}{1 + i(s)}.$$

Taking logs gives

$$\log \beta(s) = -\log(1 + i(s)).$$

Therefore, fluctuations in the interest rate map directly into fluctuations in the log discount factor.

Using a first-order approximation around the time- t interest rate i_t ,

$$d \log \beta(s) = -d \log(1 + i(s)) \simeq -\frac{1}{1 + i_t} di(s).$$

Hence, under the risk-neutral measure $\mathbb{Q}$,

$$\text{Var}_t^{\mathbb{Q}}(\log \beta_{t+h}) \simeq \frac{1}{(1 + i_t)^2} \text{Var}_t^{\mathbb{Q}}(i_{t+h}).$$

Let $\sigma_{i,t}^{IV}(h)$ denote the annualized option-implied volatility of the interest rate over horizon h . We interpret this object as a market-based measure of the risk-neutral conditional volatility of the future interest rate. Thus,

$$\text{Var}_t^{\mathbb{Q}}(i_{t+h}) \simeq h \left(\sigma_{i,t}^{IV}(h) \right)^2.$$

Substituting this expression into the previous equation gives

$$\text{Var}_t^{\mathbb{Q}}(\log \beta_{t+h}) \simeq \frac{h \left(\sigma_{i,t}^{IV}(h) \right)^2}{(1 + i_t)^2}.$$

Since

$$\widehat{\beta}_{t+h} = \log \beta_{t+h} - \log \bar{\beta},$$

subtracting the constant $\log \bar{\beta}$ does not affect the variance. Therefore,

$$\text{Var}_t^{\mathbb{Q}}(\widehat{\beta}_{t+h}) = \text{Var}_t^{\mathbb{Q}}(\log \beta_{t+h}).$$

It follows that

$$\text{Var}_t^{\mathbb{Q}} \left(\widehat{\beta}_{t+h} \right) \simeq \frac{h \left(\sigma_{i,t}^{IV}(h) \right)^2}{(1+i_t)^2} \simeq h \left(\sigma_{i,t}^{IV}(h) \right)^2.$$

The last approximation comes from $(1+i_t)^2 \simeq 1$ when i_t is small. This expression provides a direct mapping from option-implied interest-rate volatility to risk-neutral discount-factor uncertainty. The mapping does not require option-implied volatility to be identical to the physical variance of $\widehat{\beta}$. Rather, it shows that, under the model relation $\beta(s)(1+i(s)) = 1$, market-implied uncertainty about future interest rates translates into market-implied uncertainty about the stochastic discount factor.

Appendix C Construction of the Duration Measure

This section outlines the procedure for constructing cash-flow duration, following the methodologies of [Dechow, Sloan, and Soliman \(2004\)](#) and [Weber \(2018\)](#). This measure represents the weighted average timing of cash flows

$$\text{Duration}_{i,t} = \frac{\sum_{s=1}^T s \times \text{CF}_{i,t+s} / (1+r)^s}{P_{i,t}},$$

where $\text{Duration}_{i,t}$ is the duration of firm i at time t , $\text{CF}_{i,t+s}$ is the cash flow at time $t+s$, $P_{i,t}$ is the equity price, and r is the risk-free rate, which is assumed to be common across time and firms.

Equities do not have a well-defined finite maturity (T). Therefore, the duration formula is decomposed into a finite-period component and an infinite-horizon terminal value:

$$\text{Duration}_{i,t} = \frac{\sum_{s=1}^T s \times \text{CF}_{i,t+s} / (1+r)^s}{P_{i,t}} + \left(T + \frac{1+r}{r} \right) \times \frac{P_{i,t} - \sum_{s=1}^T \text{CF}_{i,t+s} / (1+r)^s}{P_{i,t}}. \quad (47)$$

Future cash flows are decomposed into two terms and are approximated using an AR(1) process:

$$\begin{aligned} \text{CF}_{i,t+s} &= E_{i,t+s} - (\text{BV}_{i,t+s} - \text{BV}_{i,t+s-1}) \\ &= \text{BV}_{i,t+s-1} \left[\frac{E_{i,t+s}}{\text{BV}_{i,t+s-1}} - \frac{\text{BV}_{i,t+s} - \text{BV}_{i,t+s-1}}{\text{BV}_{i,t+s-1}} \right]. \end{aligned} \quad (48)$$

Future return on equity $\left(\frac{E_{i,t+s}}{\text{BV}_{i,t+s-1}} \right)$ and growth in book equity $\left(\frac{\text{BV}_{i,t+s} - \text{BV}_{i,t+s-1}}{\text{BV}_{i,t+s-1}} \right)$ follow an AR(1) process with mean reversion.

$$\begin{aligned}\frac{E_{i,t+s}}{BV_{i,t+s-1}} &= (1 - \rho_1) \frac{\overline{E}}{\overline{BV}} + \rho_1 \frac{E_{i,t+s-1}}{BV_{i,t+s-2}}, \\ \frac{BV_{i,t+s} - BV_{i,t+s-1}}{BV_{i,t+s-1}} &= (1 - \rho_2) \overline{BVG} + \rho_2 \frac{BV_{i,t+s-1} - BV_{i,t+s-2}}{BV_{i,t+s-2}},\end{aligned}$$

where $\frac{\overline{E}}{\overline{BV}}$ denotes the average return on equity, and $\overline{BVG}$ denotes the average growth in book equity.⁹ After estimating each equation, the return on equity has an AR(1) coefficient of 0.67, while the growth in book equity has an AR(1) coefficient of 0.18.

Given these parameters, the procedure to calculate one-period-ahead cash flow is as follows:

1. Compute $\frac{BV_{i,t+s+1} - BV_{i,t+s}}{BV_{i,t+s}}$ using the AR(1) process.
2. Compute $\frac{E_{i,t+s+1}}{BV_{i,t+s}}$ with an AR(1) process.
3. Compute $CF_{i,t+s+1}$ using equation (48).
4. Update $BV_{i,t+s+1}$ with

$$BV_{i,t+s+1} = \left(1 + \frac{BV_{i,t+s+1} - BV_{i,t+s}}{BV_{i,t+s}} \right) BV_{i,t+s}.$$

This is a recursive procedure in which n -period-ahead cash flows are calculated in the same manner. Duration is then measured using these future cash flows and equation (47).

C.1 Derivation of Equation (5)

This subsection derives equation (5). The cash-flow duration is given by

$$\text{Duration}_{i,t} = \frac{\sum_{s=1}^{\infty} s \times CF_{i,t+s} / (1+r)^s}{P_{i,t}}.$$

This equation is decomposed into a finite term and an infinite term:

$$\text{Duration}_{i,t} = \frac{\sum_{s=1}^T s \times CF_{i,t+s} / (1+r)^s}{\sum_{s=1}^T CF_{i,t+s} / (1+r)^s} \frac{\sum_{s=1}^T CF_{i,t+s} / (1+r)^s}{P_{i,t}}$$

⁹These are set to 0.03 and 0.015, respectively, with the risk-free rate fixed at 0.03 and the terminal period T set to 60 quarters, as specified in [Weber \(2018\)](#).

$$+ \frac{\sum_{s=T+1}^{\infty} s \times \text{CF}_{i,t+s}/(1+r)^s}{\sum_{s=T+1}^{\infty} \text{CF}_{i,t+s}/(1+r)^s} \frac{\sum_{s=T+1}^{\infty} \text{CF}_{i,t+s}/(1+r)^s}{P_{i,t}}. \quad (49)$$

It is assumed that the terminal cash flow stream is equal to the difference between the observed market value implied by the stock price and the present discounted value of cash flows over the finite period,

$$\sum_{s=T+1}^{\infty} \text{CF}_{i,t+s}/(1+r)^s = P_{i,t} - \sum_{s=1}^T \text{CF}_{i,t+s}/(1+r)^s. \quad (50)$$

It is also assumed that after $t = T$ the cash flow is constant over time. Then,

$$\frac{\sum_{s=T+1}^{\infty} s \times \text{CF}_{i,t+s}/(1+r)^s}{\sum_{s=T+1}^{\infty} \text{CF}_{i,t+s}/(1+r)^s} = T + \frac{1+r}{r}.$$

By substituting this and equation (50) into equation (49), we obtain equation (5).

C.2 Negative relationship between cash-flow duration and book-to-market

Section 5.2 uses the book-to-market ratio as an alternative measure of duration for robustness checks. In this subsection, a linear relationship between cash-flow duration and the book-to-market ratio is derived under certain assumptions.

The cash-flow duration is defined as

$$\text{Duration}_{i,t} = \frac{\sum_{s=1}^T s \times \text{CF}_{i,t+s}/(1+r)^s}{P_{i,t}} + \left(T + \frac{1+r}{r}\right) \times \frac{P_{i,t} - \sum_{s=1}^T \text{CF}_{i,t+s}/(1+r)^s}{P_{i,t}},$$

and with the accounting identity, net cash distribution is given by

$$\text{CF}_{i,t+s} = \text{BV}_{i,t+s-1} \left[\frac{\text{E}_{i,t+s}}{\text{BV}_{i,t+s-1}} - \frac{\text{BV}_{i,t+s} - \text{BV}_{i,t+s-1}}{\text{BV}_{i,t+s-1}} \right].$$

If we assume that the growth in the book value of equity is zero for all finite periods ($\text{BV}_{i,t+s-1} = \text{BV}_{i,t+s}$) and the return on equity is constant across periods $\left(\frac{\text{E}_{i,t+s}}{\text{BV}_{i,t+s-1}} = \frac{\text{E}_{i,t}}{\text{BV}_{i,t-1}}\right)$, then $\text{CF}_{i,t+s} = \text{E}_{i,t}$ holds.

Under this assumption, cash-flow duration can be written as

$$\text{Duration}_{i,t} = T + \frac{1+r}{r} - T \frac{E_{i,t}}{rP_{i,t}}.$$

We also assume that the return on equity is always equal to the cost of capital, $\frac{E_{i,t}}{BV_{i,t-1}} = r$. Then, duration can be written as

$$\text{Duration}_{i,t} = T + \frac{1+r}{r} - T \frac{BV_{i,t}}{P_{i,t}}.$$

In this special case, duration is negatively and linearly related to the book-to-market ratio.

Appendix D Horse-Race Regression

A potential concern is that the interest rate uncertainty measure may not capture a distinct uncertainty channel but may simply reflect conventional monetary policy surprises. If this is the case, the relationship between changes in interest rate uncertainty and bond returns should disappear once standard high-frequency monetary policy surprises are controlled for. To address this concern, we estimate horse-race regressions that include changes in interest rate uncertainty and monetary policy surprises in the same specification. Specifically, we compare the three regressions. The first regression includes changes in interest rate uncertainty and its interaction with duration. The second regression includes monetary policy surprises and their interaction with duration. The third regression includes both variables and their interactions with duration. The coefficient of the interaction between interest rate uncertainty and duration measures whether changes in uncertainty explain the cross-duration pattern of bond returns after controlling for the conventional monetary policy surprise channel.

Tables 6 and 7 present these results. We use the principal component as the uncertainty measure in these horse-race regressions. The interaction between interest-rate uncertainty and duration remains negative and statistically significant after controlling for monetary policy surprises. This result indicates that longer-duration bonds respond more strongly to reductions in interest-rate uncertainty even conditional on standard high-frequency monetary policy surprises. In contrast, the inclusion of monetary policy surprises does not eliminate the explanatory power of the uncertainty measure. These findings suggest that the interest-rate uncertainty channel is not simply a relabeling of conventional monetary policy surprises. Instead, changes in interest-rate uncertainty contain independent information about a decline in discount-rate risk on FOMC announcement days.

Table 6: Horse-Race Regression Using Treasury Bonds.

	(1) PC only	(2) MPS only	(3) Horse race
PC	-9.51*** (2.01)		-5.04** (1.94)
Duration $\times$ PC	-0.78** (0.38)		-0.65* (0.36)
MPS		-322.44*** (46.08)	-267.48*** (49.24)
Duration $\times$ MPS		-14.54 (10.84)	-7.44 (10.38)
R ²	0.09	0.08	0.12
Observations	5,974	5,974	5,974

Note: Table 6 reports pooled regressions of Treasury bond returns on FOMC announcement days. The dependent variable is the Treasury bond return, measured in basis points. PC denotes the principal component and MPS denotes the high-frequency monetary policy surprise. Duration is measured in years. Column (1) includes PC and its interaction with duration. Column (2) includes MPS and its interaction with duration. Column (3) includes both sets of variables. Standard errors, clustered by FOMC date, are reported in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$.

Table 7: Horse-Race Regression Using Equity Returns.

	(1) PC only	(2) MPS only	(3) Horse race
MPS $\times$ Duration		-15.94* (8.11)	-3.73 (8.55)
PC $\times$ Duration	-1.22*** (0.35)		-1.15*** (0.39)
R ²	0.07	0.07	0.07
Observations	686,324	686,324	686,324

Note: Table 7 reports firm-level regressions of equity returns on FOMC announcement days. PC denotes the change in principal component, and MPS denotes the high-frequency monetary policy surprise. Duration is the equity-duration measure. Column (1) includes PC and its interaction with duration. Column (2) includes MPS and its interaction with duration. Column (3) includes both sets of variables. All specifications include firm and time fixed effects. Standard errors are clustered at the firm and time levels. Significance levels: *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$.

Appendix E Random-Assignment Placebo Tests

We conduct random-assignment placebo tests to examine whether the estimated duration coefficients arise from a chance alignment between asset returns and changes in interest-rate uncertainty.

This exercise addresses the concern that our results may be driven by a small number of influential announcement dates, generic time-series variation in uncertainty, or an accidental matching of uncertainty changes with the cross-sectional distribution of duration. If these mechanisms were sufficient to generate our findings, randomly assigned daily uncertainty changes would frequently produce duration-interaction coefficients comparable to the actual estimate.

For each placebo repetition, we randomly draw changes in interest-rate uncertainty from the full daily distribution, including both FOMC and non-FOMC days, and assign these placebo shocks to the FOMC announcement dates in our estimation sample. On each announcement date, the same placebo shock is assigned to all bond maturities or all firms. We then re-estimate the corresponding baseline specifications while leaving returns, duration measures, monetary policy surprises, controls, and fixed effects at their actual values. We repeat this procedure 5,000 times.

For each repetition, we reestimate equation (10) for bonds and equation (12) for equities. In each repetition, we record the coefficient on the interaction between the placebo uncertainty shock and duration. The randomization-based p -value is the fraction of the placebo estimates whose absolute value is at least as large as the absolute value of the coefficient estimated using an actual uncertainty shock.

The actual interaction coefficient is unusually large in absolute value relative to the placebo distribution for each asset class. The randomization-based p -value is 0.003 for bonds and 0.030 for equities. Thus, only approximately 0.3% of the placebo estimates for bonds and 3.0% of those for equities are at least as extreme as the corresponding estimates obtained using the actual interest-rate uncertainty shocks. These results indicate that the estimated duration coefficients are unlikely to reflect a coincidental assignment of daily uncertainty changes to FOMC announcement dates.

Appendix F A Cieslak–Pang Decomposition of Uncertainty Shocks

This appendix decomposes changes in equity-market uncertainty into an interest rate risk and equity-specific components. The decomposition builds on the central identification idea of [Cieslak and Pang \(2021\)](#). Cieslak and Pang use the joint behavior of stock and bond-market innovations, together with economically motivated sign restrictions to distinguish the common shocks affecting the two markets. We apply the same cross-market identification logic to measures of option-implied uncertainty.

We distinguish between a shock that jointly changes equity and interest rate risk and a shock

that primarily changes equity risk. An interest rate risk shock should move the VIX and measures of interest-rate uncertainty in the same direction. By contrast, an equity-specific risk shock should move the VIX without producing a comparably large movement in interest-rate uncertainty.

F.1 Equity and Interest-Rate Uncertainty

Figure 9 illustrates the relationship between equity and interest-rate uncertainty. It plots the daily log change in the VIX against the daily log change in interest rate uncertainty on FOMC announcement days. The solid line reports the fitted linear relationship, and the shaded area denotes its 95 percent confidence interval.

Changes in the two measures are positively related. Thus, increases in interest-rate uncertainty tend to coincide with increases in equity-market uncertainty. At the same time, the substantial dispersion around the fitted line indicates that the VIX contains considerable variation that is not shared with interest rate uncertainty. This motivates our decomposition of uncertainty into an interest rate risk component and an equity-specific component.

F.2 Identification of the Uncertainty Shocks

Following Cieslak and Pang (2021), we use the joint movements of several financial-market variables to identify economically interpretable shocks. Our daily information set is as follows:

$$x_t = \begin{pmatrix} \Delta \log VIX_t \\ \Delta \log IRU_t \\ \Delta \log BPVol_t \\ \Delta \log MOVE_t \end{pmatrix}. \quad (51)$$

The first variable measures equity-market uncertainty. The remaining three variables,

$$\Delta \log IRU_t, \quad \Delta \log BPVol_t, \quad \Delta \log MOVE_t,$$

are all measures of interest rate risk. Using three distinct measures allows the identification to exploit their common variation and prevents results from depending on any single measure of interest-rate uncertainty.

We first remove predictable movements in these variables by estimating a first-order VAR using

all business days for which the four measures are jointly available. We then summarize the common variation in the resulting daily innovations using two orthogonal factors.

As in [Cieslak and Pang \(2021\)](#), the statistical factors do not initially have a unique economic interpretation. We therefore consider rotations of the two factors and retain those satisfying economically motivated restrictions on their contemporaneous effects.

A positive *interest rate risk shock* must increase the VIX and each of the three interest rate risk measures:

$$\begin{aligned} a_{VIX,IR} &> 0, \\ a_{IRU,IR} &> 0, \quad a_{BPVol,IR} > 0, \quad a_{MOVE,IR} > 0. \end{aligned} \tag{52}$$

Accordingly, a negative realization of this shock represents a reduction in interest rate risk that lowers both equity-market uncertainty and all three measures of interest-rate uncertainty.

A positive *equity-specific risk shock* must increase the VIX while having only limited effects on the three interest rate risk measures.

$$\begin{aligned} a_{VIX,EQ} &> 0, \\ |a_{j,EQ}| &\leq \tau |a_{VIX,EQ}|, \quad j \in \{IRU, BPVol, MOVE\}. \end{aligned} \tag{53}$$

We set $\tau = 0.35$. Thus, the absolute response of each interest rate risk measure to the equity-specific shock cannot exceed 35 percent of the corresponding VIX response. A negative realization of this shock reduces equity-market uncertainty without generating a comparably large decline in interest rate risk.

These restrictions follow the set-identification approach of [Cieslak and Pang \(2021\)](#): they generally identify a collection of admissible rotations rather than a unique rotation. We draw 200,000 rotations, and retain those satisfying equations (52) and (53), and select the median-target rotation. The selected rotation is the admissible solution whose impact coefficients are jointly closest to the median impact coefficients across the identified set.

The shocks are estimated using the full daily sample. We then retain their realizations on FOMC announcement days and merge them with the stock-return sample. Consequently, the occurrence of an FOMC announcement is not used to identify the shocks.

Our procedure is motivated by considerations similar to those in the four-shock decomposition of [Cieslak and Pang \(2021\)](#). Instead, it applies their central identification principle—using restricted joint movements across financial markets—to separate the interest-rate-related and equity-specific

components of option-implied uncertainty.

F.3 Equity-Specific Risk and Equity Duration

We next examine how stocks with different equity durations respond to the identified equity-specific risk shock. To study the component of equity uncertainty that is not primarily driven by interest rate risk, we estimate an analogous regression with the identified equity-specific risk shock:

$$y_t^m = \alpha^m + \beta_{eq}^m \varepsilon_t^{EQ} + \epsilon_t^m, \quad m = 1, \dots, 10, \quad (54)$$

where ε_t^{EQ} denotes the equity-specific risk shock. The regression is estimated separately for each duration decile. Because a positive shock represents an increase in equity-specific risk, a negative coefficient implies that higher equity-specific risk lowers stock returns.

Figure 10 reports the estimates. Panel (a) sorts stocks using the cash-flow-based duration measure. The coefficients generally become more negative as cash-flow duration increases, although the relationship is not completely monotonic in the highest decile. Thus, stocks with more distant cash flows are generally more adversely affected by an increase in equity-specific risk.

Panel (b) sorts stocks by book-to-market. The estimated response is most negative among low-book-to-market stocks and becomes progressively less negative as book-to-market increases. Because low-book-to-market stocks generally have longer equity duration, this result provides complementary evidence that exposure to equity-specific risk is stronger among long-duration stocks.

Relation to the model We briefly discuss how we interpret these results through the lens of our model. As shown in Equation (3), the difference between long- and short-duration equity returns is given by

$$R^L - R^S \simeq \left(\gamma - \frac{1}{\psi} \right) \text{Cov} \left(\hat{H}, \hat{\beta} + \frac{1}{\psi} (\hat{c}_3 - \hat{c}_4) - \hat{x}_3 + \hat{x}_4 \right).$$

The cash-flow component is

$$R^L - R^S \simeq \left(\gamma - \frac{1}{\psi} \right) \text{Cov} \left(\hat{H}, \frac{1}{\psi} (\hat{x}_3 - \hat{x}_4) \right).$$

This expression implies that long-duration equities earn a higher return if and only if

$$\left(\gamma - \frac{1}{\psi}\right) \text{Cov}\left(\hat{H}, \frac{1}{\psi}(\hat{x}_3 - \hat{x}_4)\right) > 0.$$

Under the reasonable assumption that $\gamma > \frac{1}{\psi}$, this condition becomes

$$\text{Cov}\left(\hat{H}, \frac{1}{\psi}\hat{x}_3\right) > \text{Cov}\left(\hat{H}, \frac{1}{\psi}\hat{x}_4\right).$$

Economically, this condition means that short-horizon cash flows have a larger covariance with the announcement pricing kernel than long-horizon cash flows. One way to interpret this condition is that near-term cash flows are relatively predetermined and therefore less sensitive to adverse aggregate states, whereas distant cash flows may depend more strongly on long-run growth opportunities and profitability.

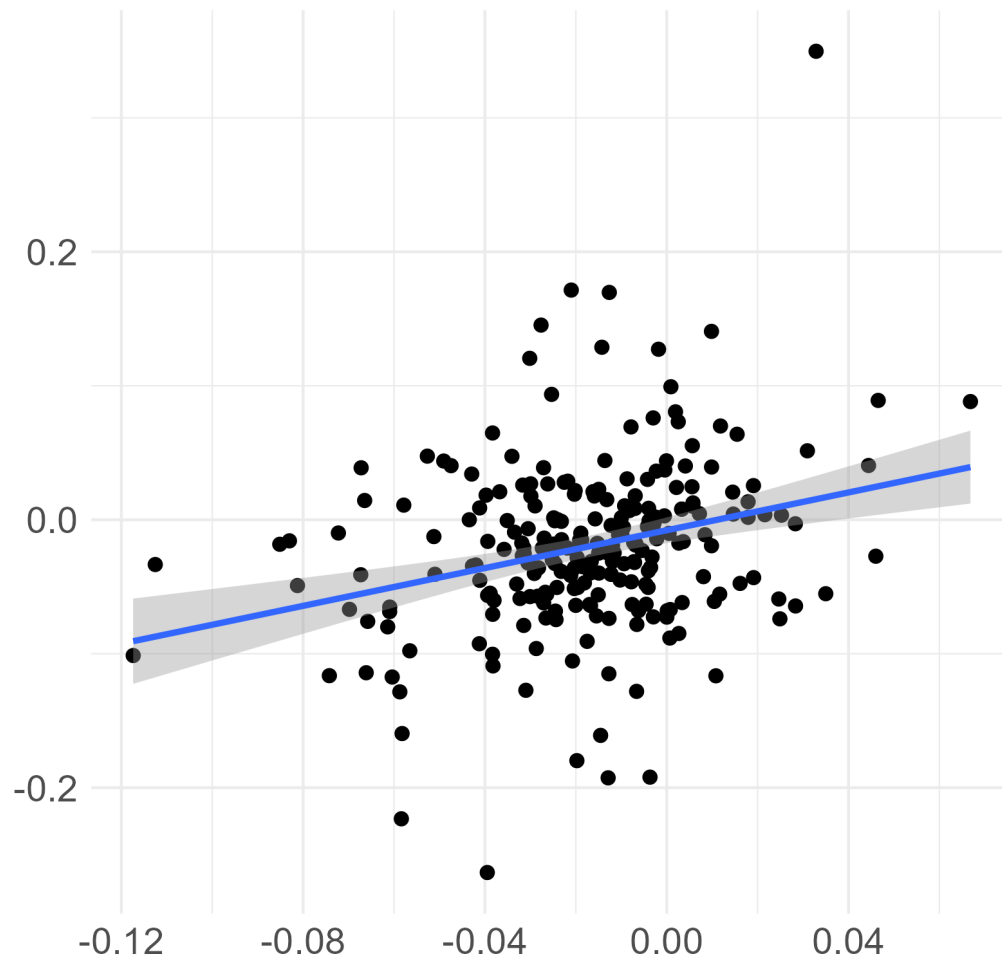
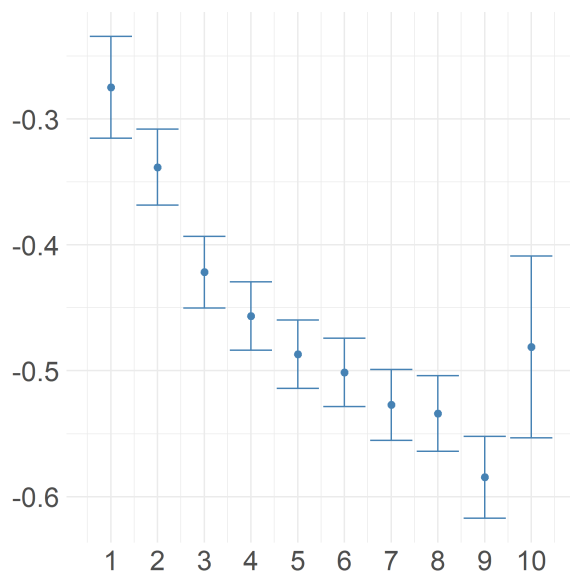
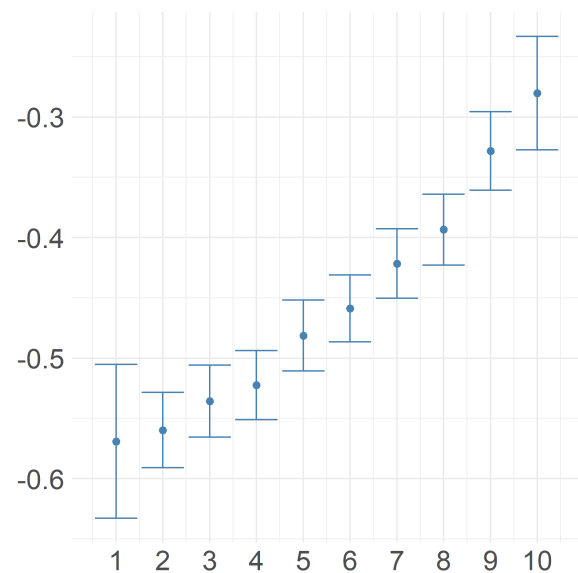


Figure 9: **VIX and Interest Rate Uncertainty.** The figure plots the daily log change in the VIX against the daily log change in interest rate uncertainty on FOMC announcement days. The solid line reports the fitted linear relationship, and the shaded area denotes the 95 percent confidence interval.



(a) Cash-flow-duration deciles



(b) Book-to-market deciles

Figure 10: Equity-specific Risk and Equity Duration. The figure reports coefficients from separate regressions of FOMC announcement-day excess returns on the identified equity-specific risk shock. Panel (a) sorts stocks into deciles using the cash-flow-based duration measure. Panel (b) sorts stocks into deciles using book-to-market. Error bars denote the point estimate with two standard error bands.

Table 8: Bond Return Sensitivity to Interest Rate Risk: Alternative Sample Periods.

	(1)	(2)
IRU	-1.29 (0.91)	-3.13** (1.43)
Duration $\times$ IRU	-0.47* (0.25)	-0.44** (0.20)
MP shock	-259.56*** (33.80)	-287.77* (150.52)
Duration $\times$ MP shock	-1.35 (9.20)	-27.53 (23.90)
R ²	0.16	0.23
Observations	3,248	2,726

Note: This table reports coefficient estimates from pooled regressions of Treasury bond returns on changes in interest rate uncertainty, duration, and their interaction. Standard errors are clustered by FOMC date. The dependent variable is the Treasury bond return, expressed in basis points. Column (1) uses the pre-crisis period (1994–2007); column (2) uses the post-crisis period (2008–2019). All specifications include the daily change in VIX, the monetary policy shock, the interaction between the change in VIX and duration, and the interaction between the monetary policy shock and duration as controls. Significance levels: *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$.

Appendix G Sensitivity Analysis for Bonds

G.1 Alternative Sample Periods

We also conduct subsample analyses to examine whether the results are sensitive to the sample period. We divide the sample into the pre-crisis period (1994–2007) and the post-crisis period (2008–2019). The empirical specification follows equation (10), with the uncertainty measure given by the benchmark interest rate risk variable, ΔIRU_t . The results are reported in Table 8. Columns (1) and (2) report the results for the pre-crisis and post-crisis periods, respectively. Across both subsamples, the interaction term between duration and the uncertainty measure is negative.

Appendix H Sensitivity Analysis for Equities

H.1 Sensitivity to Alternative Sample Periods

H.1.1 Average Returns on Equities

Figure 11 presents the average returns of portfolios sorted by cash-flow duration. Panel (a) uses the 1994–2007 sample, panel (b) uses the 2008–2019 sample.

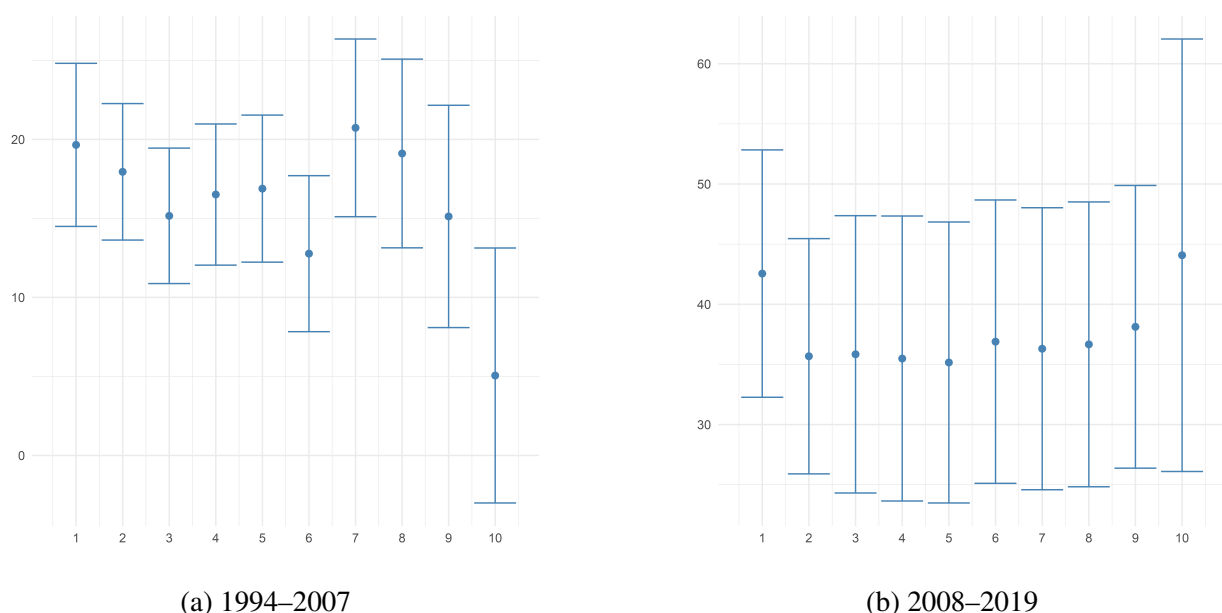


Figure 11: Average Returns Conditional on Duration: Different Time Subsamples

Note: This figure plots the time-series average of equity portfolio returns on FOMC announcement days across different subsamples. The horizontal axis represents portfolio duration, ranging from short to long duration. The vertical axis represents the average return on FOMC announcement days for each portfolio. Equities are sorted into ten portfolios based on duration, and the average return within each portfolio is calculated. Portfolios are rebalanced every quarter. Panel (a) uses the pre-crisis sample 1994–2007, panel (b) uses the post-crisis sample 2008–2019.

H.1.2 The Sensitivity of Returns to Interest Rate Risk

Table 9 estimates regression (12) using different time subsamples. Columns (1) and (2) use the 1994–2007 sample, while columns (3) and (4) use the 2008–2019 sample. Across both subsamples, the interaction between interest rate uncertainty and duration is negative and statistically significant.

Table 9: The Sensitivity of Returns to Interest Rate Uncertainty Conditional on Duration: Different Time Subsamples

	(1)	(2)	(3)	(4)
IRU $\times$ Duration	-0.38*** (0.09)	-0.50*** (0.10)	-0.51** (0.20)	-0.62*** (0.18)
R ²	0.00	0.03	0.02	0.07
Observations	440,886	391,750	245,438	231,323
Controls		✓		✓

Note: This table reports coefficient estimates from pooled regressions of individual stock returns on FOMC announcement days. The main explanatory variable is the interaction between interest rate uncertainty and equity duration. Columns (1) and (2) use the 1994–2007 sample, while columns (3) and (4) use the 2008–2019 sample. Columns (2) and (4) include firm and year fixed effects. Standard errors are clustered at the firm and time levels. Significance levels: *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$.

H.2 Sensitivity to Alternative Duration Measures

As an additional robustness check, we replace our baseline measure of equity duration with the duration measure proposed by [Gormsen and Lazarus \(2023\)](#). Column (1) uses our baseline interest-rate uncertainty shock, whereas column (2) uses the principal component shock constructed from multiple measures of interest-rate uncertainty. In both specifications, the interaction between the uncertainty shock and equity duration is negative and statistically significant. Thus, the finding that long-duration stocks respond more negatively to increases in interest-rate uncertainty is robust to both an alternative measure of equity duration and an alternative measure of the uncertainty shock.

H.3 Sensitivity to Parameters in Cash-Flow Duration

To construct the cash-flow duration, the required parameters include the persistence of ROE (ρ_1), the persistence of book-equity growth (ρ_2), the long-run growth rate of book equity ($\overline{BVG}$), the long-run mean of ROE ($\overline{\frac{E}{BV}}$), the discount rate (r), and the forecasting horizon (T). All variables are defined in Appendix C.

We conduct a sensitivity analysis by varying the parameters. The baseline parameter values are $\rho_1 = 0.67$, $\rho_2 = 0.18$, $\overline{BVG} = 0.015$, $\overline{\frac{E}{BV}} = 0.03$, $r = 0.03$, and $T = 60$ quarters (15 years). One parameter is changed at a time while the others are held constant. Specifically, the persistence of ROE is set to 0.55 and 0.75 (baseline 0.67); the persistence of book-equity growth to 0.1 and 0.3 (baseline 0.18); the long-run growth rate of book equity to 0.01 and 0.02 (baseline 0.015); the long-run mean of ROE to 0.02 and 0.04 (baseline 0.03); the discount rate to 0.025 and 0.035 (baseline 0.03); and the forecasting horizon to 12 years and 18 years (baseline 15 years).

Table 10: The Sensitivity of Returns to Interest Rate Uncertainty Conditional on Duration: Alternative Duration Measure

	(1)	(2)
G-L measure $\times$ IRU	-9.09*** (2.06)	
G-L measure $\times$ PC		-21.02** (9.31)
G-L measure	-16.99*** (5.39)	-40.35*** (15.34)
R ²	0.05	0.07
Observations	633,870	633,870

Note: This table reports coefficient estimates from pooled regressions of individual stock returns on FOMC announcement days. Equity duration is measured using the duration measure proposed by [Gormsen and Lazarus \(2023\)](#). Column (1) uses the baseline interest-rate uncertainty shock, whereas column (2) uses the principal component shock constructed from multiple measures of interest-rate uncertainty. The main explanatory variable is the interaction between the corresponding uncertainty shock and equity duration. Both specifications include the indicated controls. Standard errors are clustered at the firm and time levels. Significance levels: *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$.

H.3.1 Average Returns on FOMC Days

We calculate the cash-flow duration for each firm under twelve alternative parameter values. Firms are then sorted by duration, and we estimate the average returns on FOMC days for each portfolio. Figures 12 and 13 plot the average return against cash-flow duration under twelve different parameter values. In each figure, one parameter is varied while the others are held fixed at their baseline values.

H.3.2 The Contemporaneous Relationship between Interest Rate Uncertainty and Equity Returns

This subsection presents a sensitivity analysis of the sensitivity of equity returns to interest rate risk. After calculating cash-flow duration under different parameter values and sorting firms by duration, we regress portfolio returns on changes in interest rate risk.

Figures 14 and 15 show the coefficients β^m for each portfolio m along with their two-standard-error bands. The figures demonstrate that the sensitivity of equity returns to interest rate risk increases with duration in absolute value. This result does not depend on the parameter values assumed when calculating cash-flow duration.

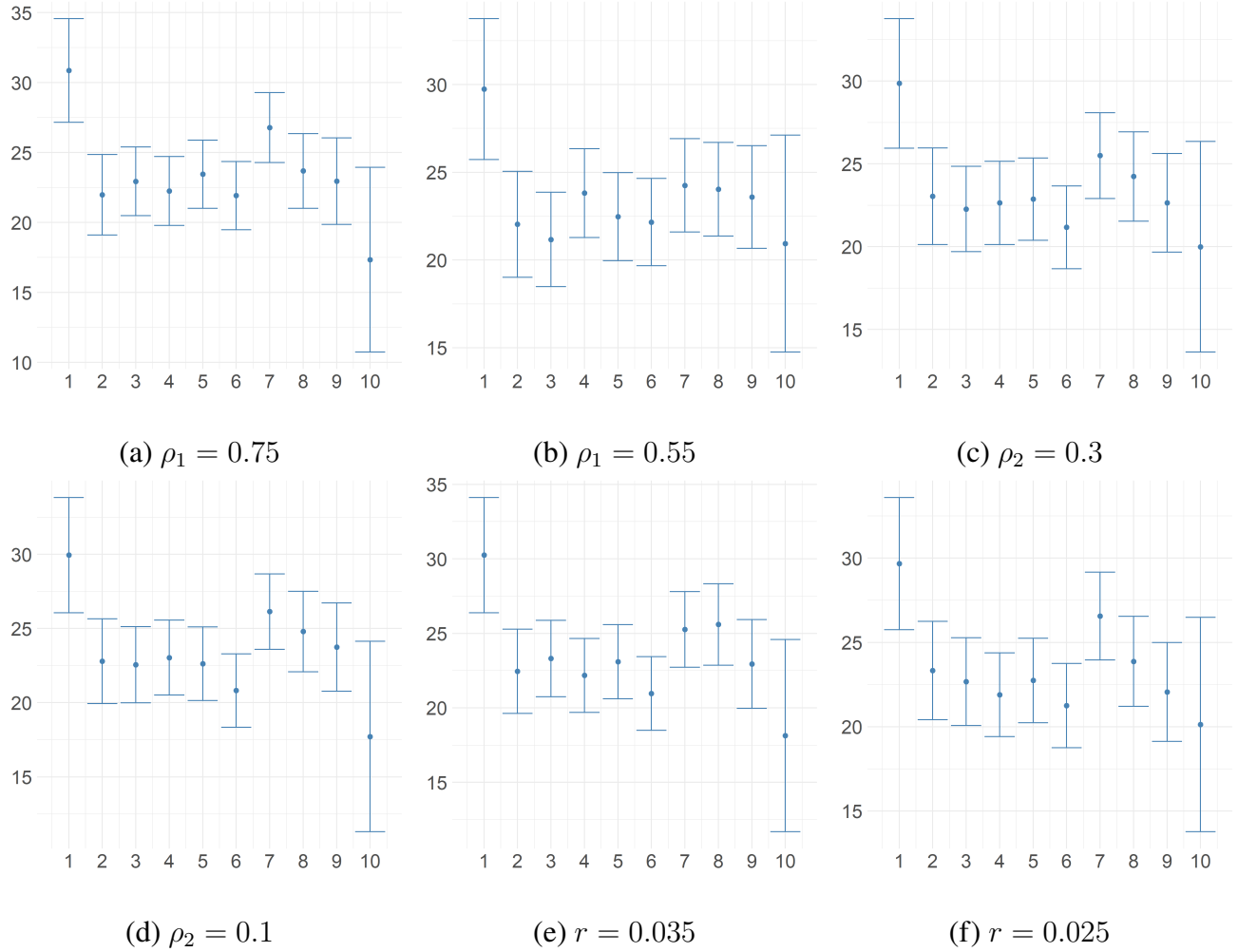


Figure 12: Sensitivity Analysis for Average Returns.

Note: Figure 12 plots the time-series average of portfolio returns on FOMC days under different parameter values. The horizontal axis represents portfolio duration, ranging from short (one) to long (ten). The vertical axis represents the average return on FOMC days for each portfolio. Equities are sorted into ten groups, from low to high, based on duration, and the average return within each group is calculated. Each panel varies one parameter at a time while holding the others constant.

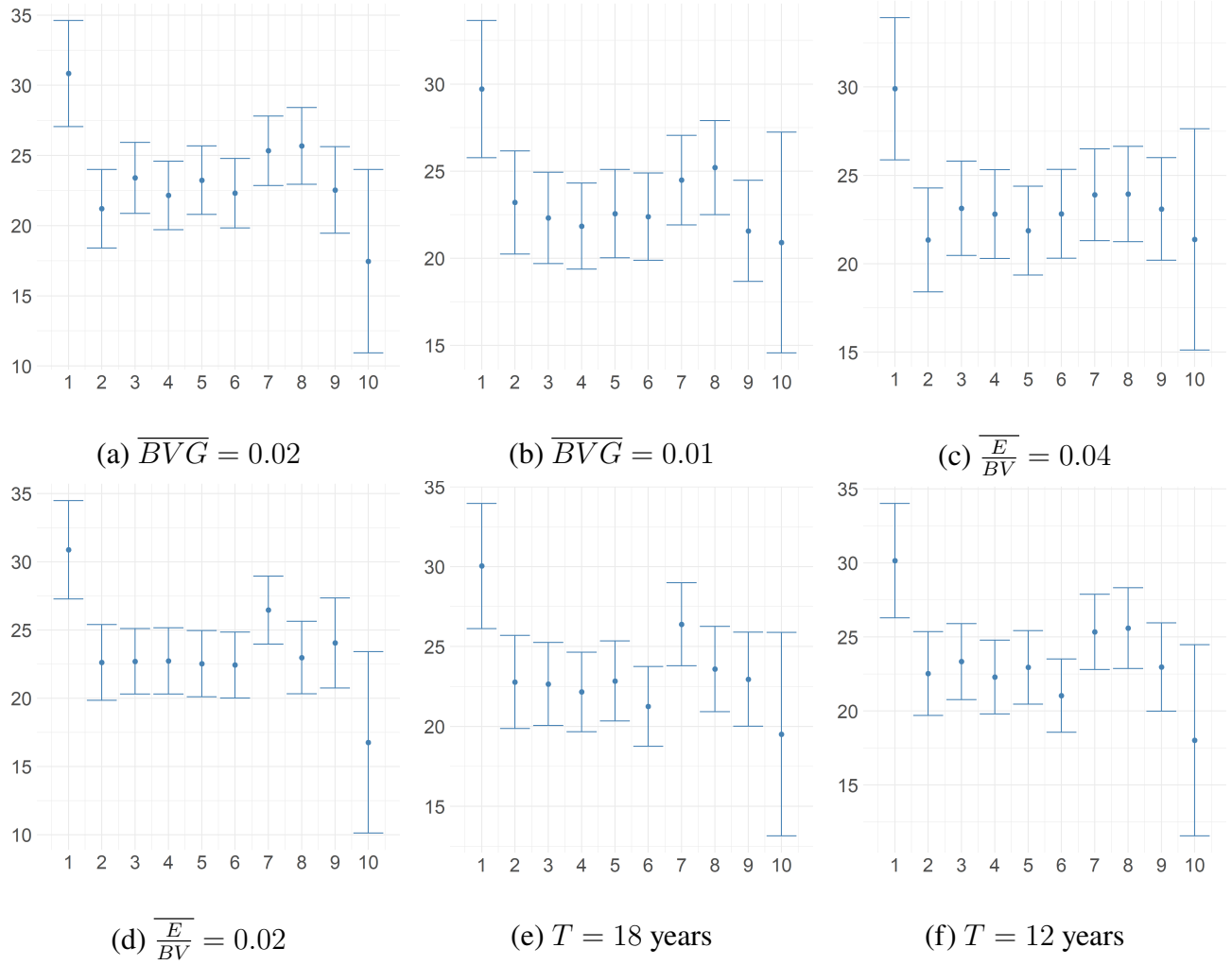


Figure 13: Sensitivity Analysis for Average Returns.

Note: Figure 13 plots the time-series average of portfolio returns on FOMC days under different parameter values. The horizontal axis represents portfolio duration, ranging from short (one) to long (ten). The vertical axis represents the average return on FOMC days for each portfolio. Equities are sorted into ten groups, from low to high, based on duration, and the average return within each group is calculated. Each panel varies one parameter at a time while holding the others constant.

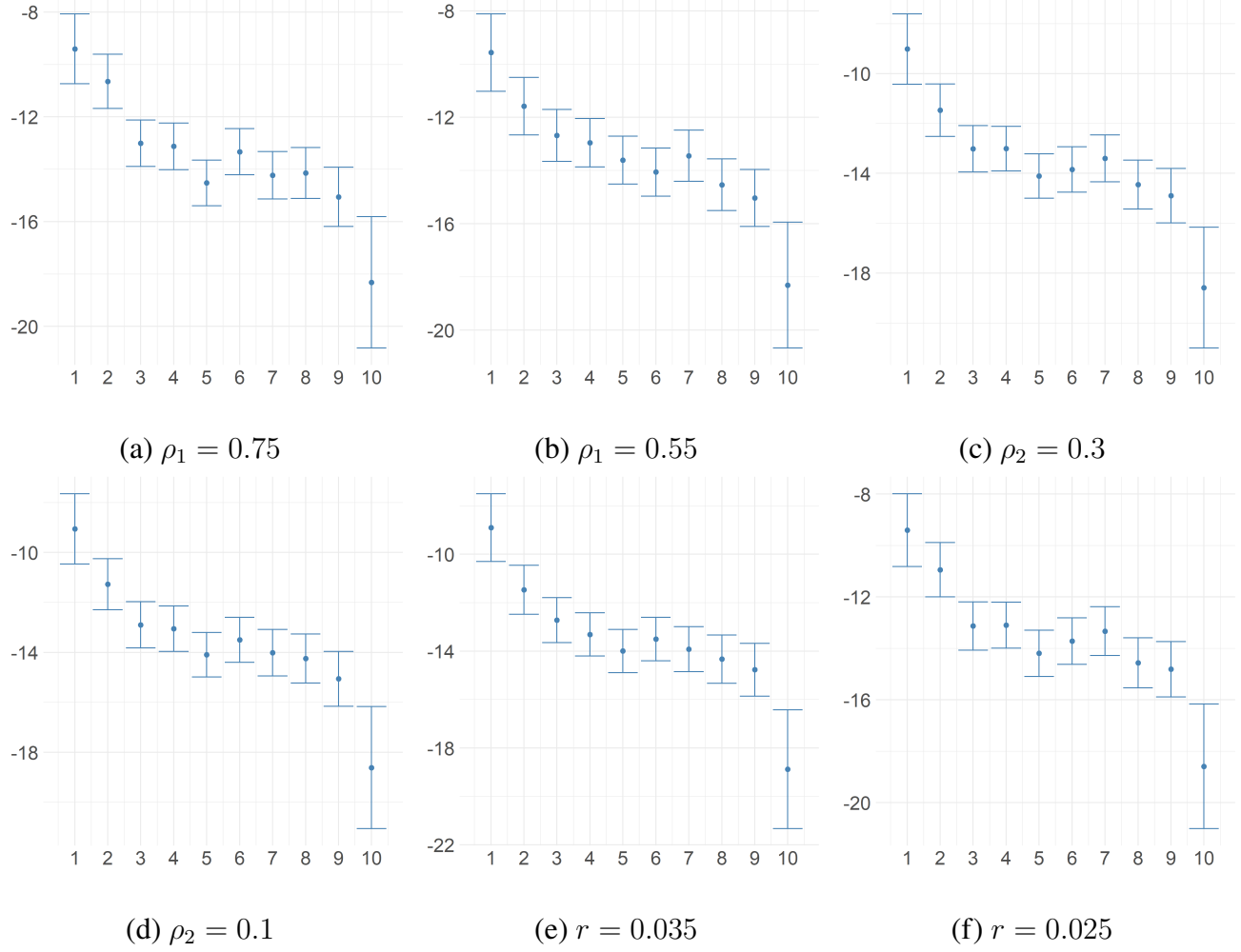


Figure 14: Robustness for the Sensitivity to Interest Rate Risk.

Note: Figure 14 shows the sensitivity of equity returns to changes in interest rate risk. We separately regress portfolio returns on changes in the uncertainty measure. The interest rate risk measure is taken from [Bauer, Lakdawala, and Mueller \(2022\)](#). The vertical axis plots the coefficients β^m for each portfolio m along with their two-standard-error bands. In each panel, one parameter is varied while the others are held constant.

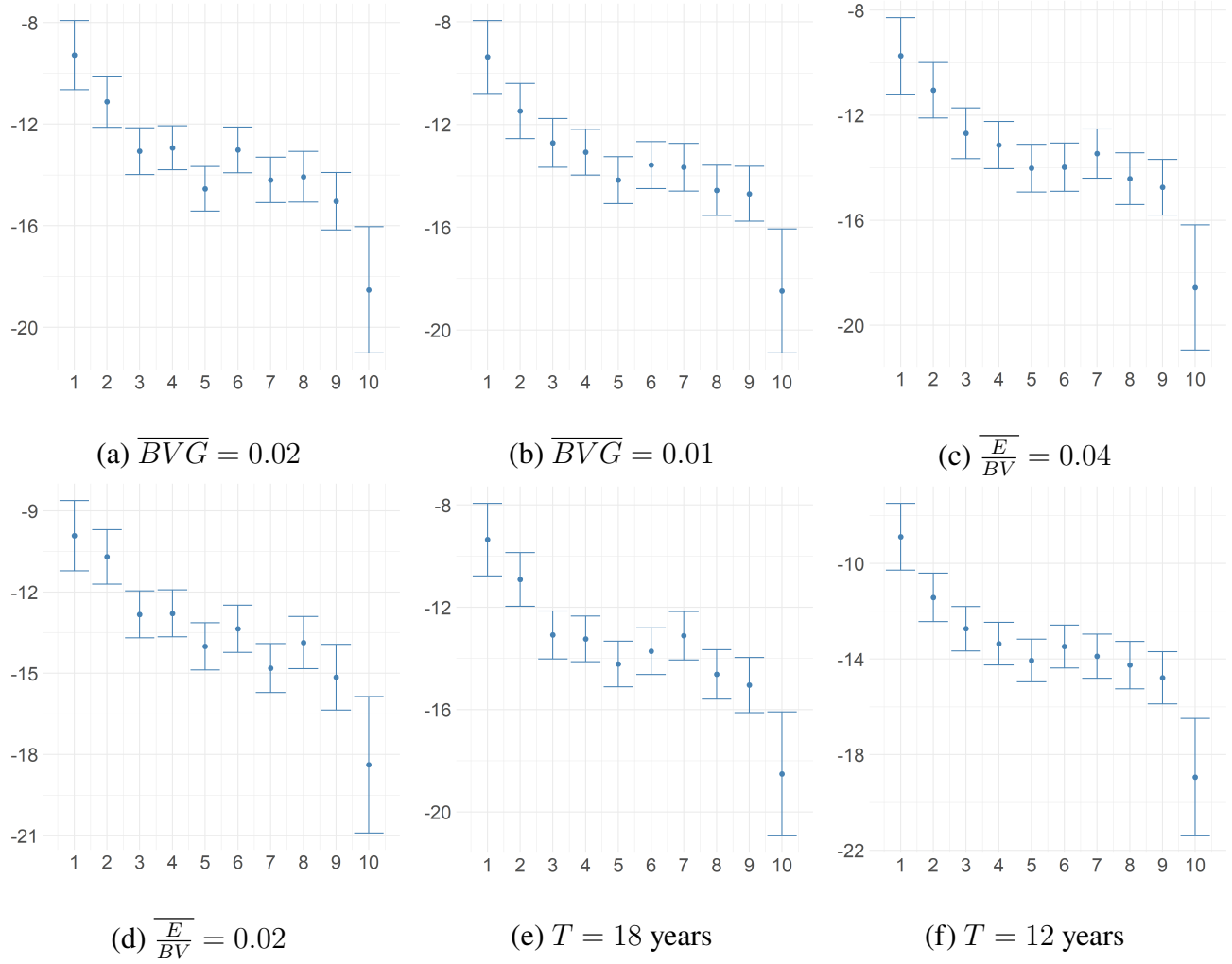


Figure 15: Robustness for the Sensitivity to Interest Rate Risk.

Note: Figure 15 shows the sensitivity of equity returns to changes in interest rate risk. We separately regress portfolio returns on changes in the uncertainty measure. The interest rate risk measure is taken from [Bauer, Lakdawala, and Mueller \(2022\)](#). The vertical axis plots the coefficients β^m for each portfolio m along with their two-standard-error bands. Each panel varies one parameter at a time while holding the others constant.

H.4 Further Evidence from Short-Term Dividend Strips

Measuring the duration of equities is subject to several concerns, including measurement errors and limited coverage. We therefore provide complementary evidence from different perspectives. Specifically, we use the short-term equity claims on near-term dividends of the aggregate stock market constructed by [Golez and Matthies \(2025\)](#). Their dataset provides FOMC announcement returns on 180-day and 360-day S&P 500 dividend strips, inferred from SPX options. Because these assets are claims on near-term dividends, they have shorter durations and are less exposed to discount factor risks than the aggregate equity market. We compare their announcement returns with those of the S&P 500 index. The dividend-strip sample covers scheduled FOMC announcements from January 2004 to December 2019.

Table 11 reports separate regressions of daily short-term dividend returns and S&P 500 returns on IRU and the monetary policy surprise. This exercise examines whether interest rate uncertainty affects short-horizon dividend claims and the aggregate stock market similarly. Columns (1) and (2) use the 180-day and 360-day dividend-strip returns, respectively, while Column (3) uses the S&P 500 return. The coefficients on interest rate risk are small and statistically insignificant in both dividend-strip regressions, whereas the coefficient for the S&P 500 is negative and statistically significant. These results suggest that short-horizon dividend claims do not respond significantly to changes in interest rate risk, whereas the long-duration aggregate S&P 500 responds strongly. These findings provide additional support for our proposed mechanism. A positive coefficient of the monetary policy surprise in short-term dividend-return regressions is reported in [Golez and Matthies \(2025\)](#).

Table 11: Separate Regressions for Short-Term Dividend Returns and S&P 500 Returns

	Short-term div. 180-day	Short-term div. 360-day	S&P 500
Constant	23.04 (35.01)	2.07 (13.54)	2.75 (11.32)
IRU	1.02 (5.76)	-0.54 (3.18)	-13.02*** (3.79)
Monetary policy shock	24.15*** (5.62)	7.92*** (2.60)	-6.43** (2.84)
R ²	0.10	0.07	0.25
Observations	128	128	128

Note: This table reports estimates from separate regressions of asset returns around FOMC announcements. The dependent variable is the announcement return, measured in basis points. Columns (1) and (2) use short-term S&P 500 dividend strip returns from [Golez and Matthies \(2025\)](#). Column (1) uses the 180-day short-term dividend strip return, and column (2) uses the 360-day short-term dividend strip return. Column (3) uses the S&P 500 index return. Standard errors are clustered at the time-level. Significance levels are denoted by *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$.

Table 12: Cross-sectional Heterogeneity in Average Daily Returns on FOMC and Non-FOMC Days.

	S&P 500	Group				
	index	1	2	3	4	5
Mean on FOMC	26.0	5.6	20.2	29.8	41.5	65.8
Mean on Non-FOMC	3.2	5.2	5.4	5.5	6.0	7.2
SD on FOMC	121	186	200	239	268	335
SD on Non-FOMC	117	192	206	230	253	297

Note: Table 12 shows the cross-sectional heterogeneity in average returns for S&P 500 firms. Firms are sorted into five groups based on their time-series average returns on FOMC days. The average daily returns on FOMC and non-FOMC days are calculated for each group. “Mean on FOMC” represents the time-series average return for each group on FOMC days. “Mean on Non-FOMC” represents the corresponding return on non-FOMC days. “SD” denotes the standard deviation. “S&P 500 index” represents the mean and standard deviation of the S&P 500 index. Returns are expressed in basis points. Groups “1” through “5” denote portfolios from low to high. The sample period is from January 1994 to December 2019.

Appendix I Heterogeneous Exposure to Monetary Policy Announcements

This section documents the cross-sectional heterogeneity in exposure to monetary policy announcements among S&P 500 firms. We calculate the time-series average daily returns on FOMC days for each firm over the sample period (January 1994–December 2019). Firms are then sorted into five groups, from the lowest to the highest average returns. Table 12 reports the mean and standard deviation of daily returns on both FOMC and non-FOMC days for each group. The results reveal substantial variation in FOMC-day returns across the five groups, with the highest group averaging 65.8 basis points and the lowest group averaging 5.6 basis points. In contrast, returns on non-FOMC days are similar across all groups.

This pattern complements the aggregate evidence in [Savor and Wilson \(2013\)](#), who show that the S&P 500 index earns higher returns on FOMC days than on non-FOMC days. Table 12 shows that this aggregate announcement premium is not uniformly distributed across firms. Instead, the elevated aggregate return on FOMC days is largely attributable to firms that respond strongly to monetary policy announcements.